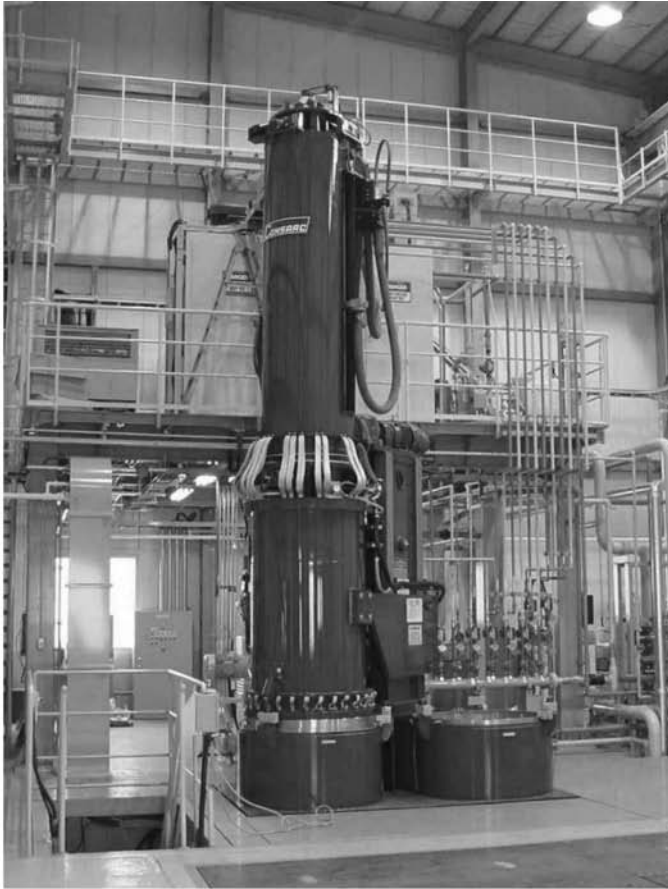
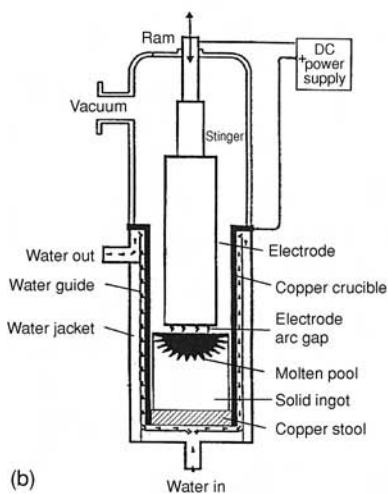
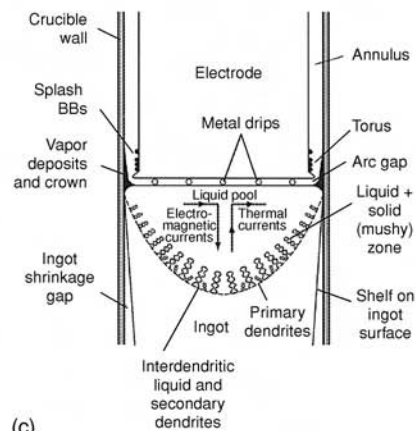




(a)



(b)



(C)

Fig. 4.6. (a) Industrial-scale vacuum arc remelting (VAR) furnace (courtesy of Jai Narayan, Consarc Corporation), (b) Schematic illustration of the VAR furnace. (c) Close-up of the details in the vicinity of the molten pool.

Table 4.1. *Chemical analysis of the bulk and the defect area in wt%*

Element	Alloy	White spot
Al	0.49	0.41
Si	0.20	0.19
Ti	0.81	0.62
Cr	17.50	17.71
Fe	17.75	19.21
Ni	54.97	55.70
Nb	5.10	2.96
Mo	3.18	3.20
T_{solidus}	1242 °C	1309 °C
T_{liquidus}	1348 °C	1373 °C

Courtesy of Laurence Jackman

been developed to enable computerised control of the melt rate, which is typically in the range 2–6 kg/min. As the electrode is melted, a liquid pool is produced which resides on the already solidified material; the size and shape of the molten zone remains approximately constant during processing, apart from brief transients associated with start-up and finishing, i.e. a steady-state is achieved. For a given electrode diameter, the depth of the pool increases with the power input, the product of the current and the applied voltage; modelling work [6] indicates that the efficiency of this process is in the range 50–75%, and that the current enters the ingot at the meniscus and at the stool at the bottom of the copper crucible on which the solidified material sits. The molten pool is subjected to complex flow patterns, which are due to a combination of the buoyancy forces, electromagnetic stirring and the Marangoni effect arising from the temperature dependence of the surface tension. The velocity of the molten material on the surface can reach 0.01 m/s; the flow pattern is radial, consistent with the symmetry of the situation, and outward from the centre of the pool; see Figure 4.7.

The resulting ingot has a structure and chemistry far superior to the electrode being melted. It will have reduced amounts of tramp elements, such as lead and bismuth, which have substantial vapour pressures and are therefore volatilised. Degassing of oxygen and nitrogen also occurs through the vacuum system, although it is difficult to remove them completely when nitride-forming elements such as titanium and niobium are present. Oxides are removed by flotation and thermal dissociation; the former is generally more effective, with surface tension and Marangoni forces playing a part in agglomerating the oxide debris. On account of the enhanced cleanliness produced, VAR remains the most important and widely used of the remelting practices for the superalloys.

Case study. Melt-related ‘white spot’ defects

Defects known as *white spots* are solute-lean regions which lack the hardening phases γ' and γ'' and the associated hardening elements such as Al, Ti and Nb [7,8]. The terminology comes from the metallographical etching techniques which provide preferential attack, leaving the solute-lean regions bright or white; see Figure 4.8 and Table 4.1. Experiments have shown that mechanical properties, such as the yield stress and ultimate tensile strength

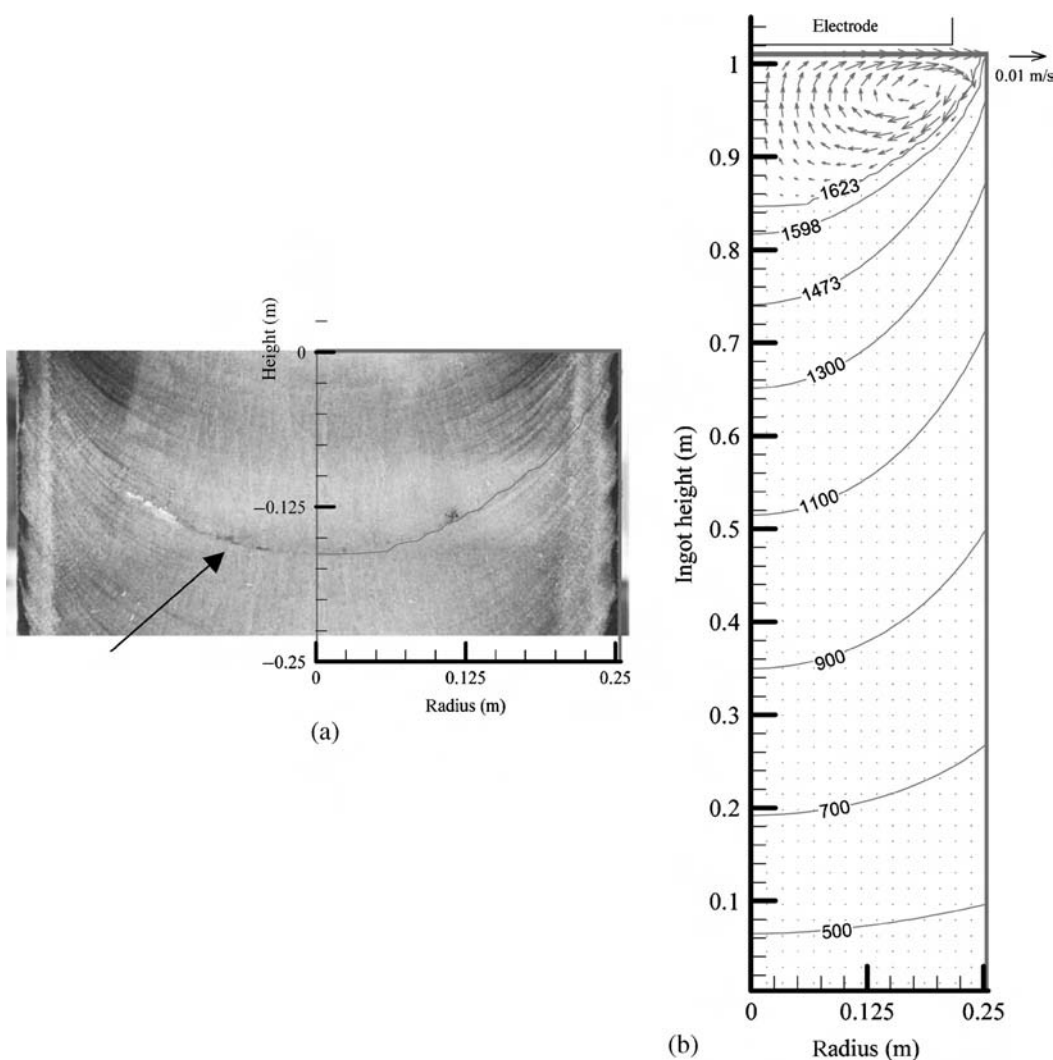
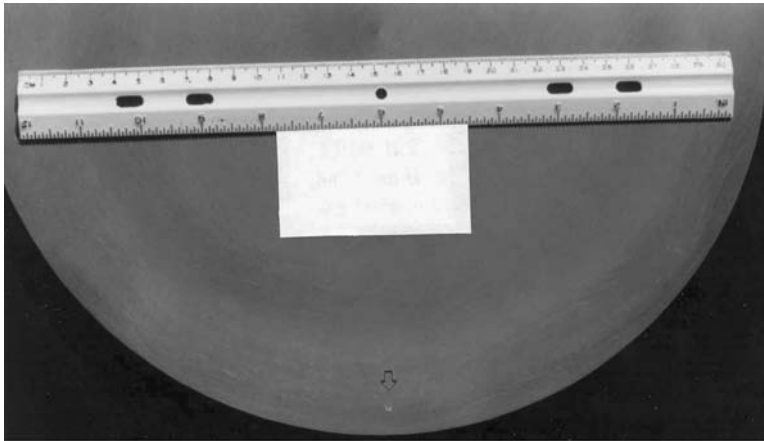


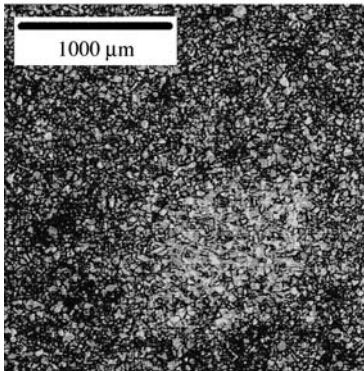
Fig. 4.7. Results from the modelling of the VAR processing of a 0.5 m diameter ingot of IN718 superalloy: (a) comparison between predicted and measured pool profile; (b) computed temperature distribution in the ingot and the flow in the molten pool [6].

(UTS), are reduced when white spots are present, and this is obviously a cause for concern. The VAR process is very much more prone to white spot than the ESR process, described below. It should be noted that there is no known method for assuring the complete absence of white spot from superalloy ingots for which VAR is the final remelting process. However, careful attention to process control for both VIM electrode production and VAR processing reduces the probability of its occurrence.

Various mechanisms have been proposed for the formation of white spot, and it seems probable that each occurs to some degree. However, a common feature is the falling of material into the molten pool, such that its differing chemistry allows it to be incorporated



(a)



(b)

Fig. 4.8. White-spot defect in the IN718 superalloy: (a) its appearance on a cross-section of a macroetched billet; (b) optical micrograph. (Courtesy of Laurence Jackman.)

into the solidifying structure without it becoming remelted. This eventuality is more likely for (i) solute-lean material with a melting temperature that is significantly larger than for the bulk composition, (ii) larger particles which do not dissolve as readily as smaller ones and (iii) larger local solidification times. Hence the frequency of white-spot occurrence is dependent upon the resistance of the process to instabilities in the process.

Where does this material originate from? Studies have shown that one source is the crown and shelf region, which occurs in the VAR process at the pool/crucible interface. During processing, small changes in current distribution cause the centre of the pool to shift, such that the remelting of some fraction of the crown is promoted; this is then undercut and material is free to drop into the molten pool. A second possibility is fragmented dendritic material from the electrode itself: obviously the interdendritic regions will melt preferentially because of their lower melting point; this emphasises the importance of the mechanical integrity of the electrode. Modelling studies [9] indicate that particles falling from the electrode are likely to melt completely before entrapment in the mushy zone,

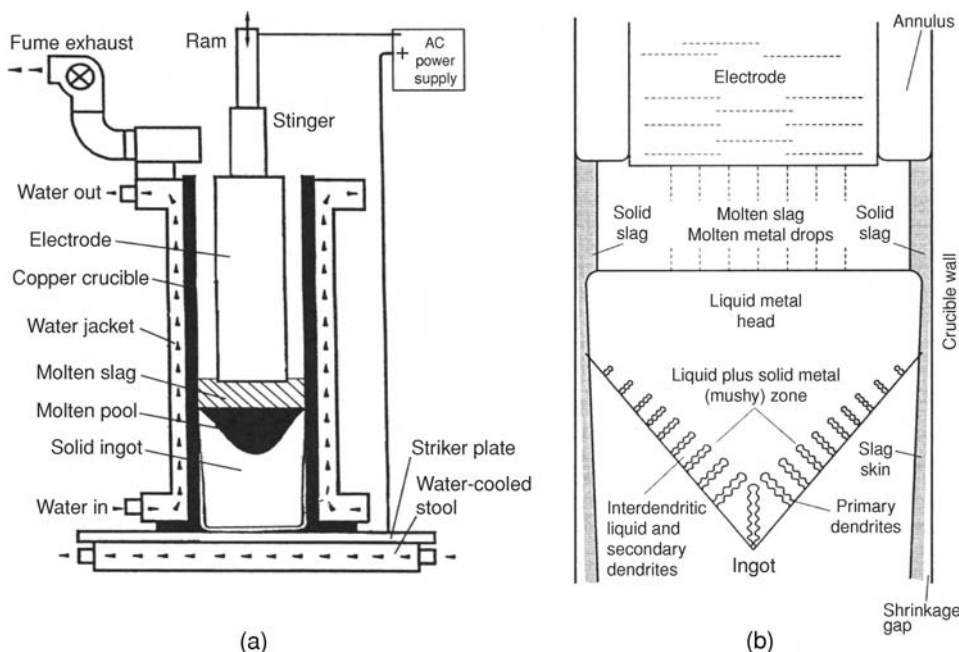


Fig. 4.9. (a) Schematic illustration of the apparatus used for the electro-slag remelting of superalloys. (b) Close-up of the details in the vicinity of the molten pool.

provided that their diameter is greater than 6 mm; crown fall-in was considered to be a more likely source of white spot, since the maximum safe diameter of material falling into the pool was found to be in the range 2 to 3 mm.

B Electro-slag remelting

Electro-slag remelting (ESR) is used widely as a secondary melting process for the production of the material for turbine discs; see Figure 4.9. It has similarities to the VAR process in that a consumable electrode is remelted, but here a molten slag pool sits between the electrode and the solidifying ingot – hence processing can be carried out in air. A major significant advantage over VAR processing then arises: molten droplets from the electrode pass through the slag, allowing them to react with it such that impurities such as oxides and sulphur are removed. However, unlike the VAR process, an alternating current (AC) is applied; the electrode immersion plus the slag cap must be regarded as the major resistance in the circuit; consequently, the majority of the heat generation occurs in this region. Changing the weight of slag in the system alters the resistance of the circuit and thus the current and voltage required to maintain a given power input/melt rate.

A number of processing controls are used which aid in the production of ingots of high quality and consistency. First, a constant immersion of the electrode in the slag is maintained, via control of the electrode voltage. Second, one monitors the rate at which the electrode melts by measuring the loss of electrode weight during processing, and adjusts the current accordingly. During solidification, slag may be added or removed as processing proceeds

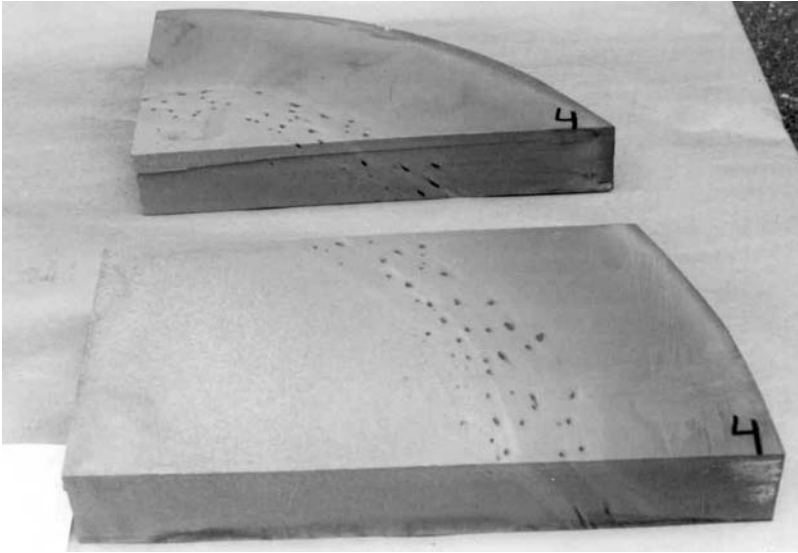


Fig. 4.10. Sections of ESR-melted superalloy billet, showing extensive evidence of the freckle defect. (Courtesy of Laurence Blackman.)

to maintain a uniform head over the pool. The choice of slag is critical, since it determines the reactivity with the molten droplets via different melting temperatures and viscosities. The slags are normally based upon fluorite CaF_2 with additions of CaO , MgO and Al_2O_3 . Impurities of SiO_2 can cause problems, since Ti reduces it to metallic Si which will be incorporated into the ingot. This is managed by recycling the slag, which then approaches an equilibrium with Ti and Si pickup, thus minimising elemental exchange. In practice, minor changes in alloy chemistry are caused in this way; ESR is much more prone to this effect than VAR.

One disadvantage of the molten slag is that it acts as a heat reservoir at the top of the ingot/pool boundary and this makes the fusion profile obtained from the ESR process different from that obtained from VAR; generally with steeper sides, larger mushy zones and decreased cooling rates for comparable power input. These factors promote levels of chemical heterogeneity that are greater in ESR than in VAR processing; hence where triple melting is employed, ESR will always precede VAR. Additionally, the maximum ingot diameter which can be melted by ESR is dependent upon alloy composition. This is because large diameters promote a greater mushy zone and enhanced local solidification times, due to the increased distances. For this reason the processing of higher-strength alloys such as U720Li is limited to diameters of about 500 mm or 20 inches. Attempts have been made [10] to use ESR processing for the production of very large 890 mm (35 inch) IN718 ingots of weight approaching 14 tonnes, each of which is sufficient for just two turbine discs for a large land-based turbine. It was reported that no conditions could be found which yielded ingots that were free of extensive freckling; the heavy freckle channels were found to be rich in elements such as Nb, Mo and Ti, which are known to partition to the liquid phase. Subsequent VAR processing was used to produce sound triple-melted (VAR + ESR +

VAR) 760 mm (30 inch) product for this application. Figure 4.10 illustrates the formation of freckle defects in ESR-processed ingots of IN718; their occurrence seems to be most likely at the mid-radius position, where the permeability of the mushy zone is greater than at the centre of the ingot and is deep enough to support the interdendritic fluid flow required for their formation [11].

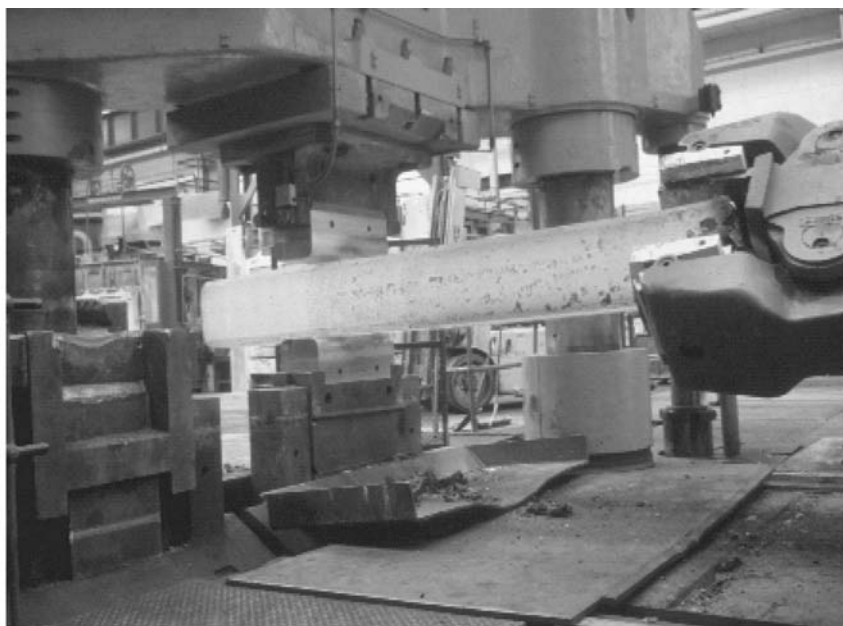
C Ingot-to-billet conversion: the cogging process

The ingots produced by the remelting processes are unsuitable for mechanical applications: they must undergo thermal-mechanical working in order to break down the as-cast structure and to reduce the grain size to acceptable levels, i.e. from a few tens of millimetres to a few tens of microns. This is known as ingot conversion and it is usually achieved by a process known as *cogging*, during which the diameter of the cylindrical ingot is reduced in size by a factor of approximately 2, so that its length increases four-fold, see Figure 4.11. The conversion is achieved in a series of stages, or heats. The ingot is placed in a furnace and allowed to reach an appropriate temperature for forging. Upon removal, it is manipulated between two horizontal dies which are driven hydraulically. These squeeze (or bite) the material, causing it to deform both in and out of the plane of the dies. Typically the ingot is deformed 20 to 30 times at various points along its length, with the pattern repeated with the ingot rotated through 90°, 45° and one final turn of 90°. At the end of this procedure, the ingot is returned to the furnace now with a characteristic octagonal cross-section. The deformation applied to the ingot causes substantial recrystallisation to a finer grain structure. After reheating the whole process is repeated.

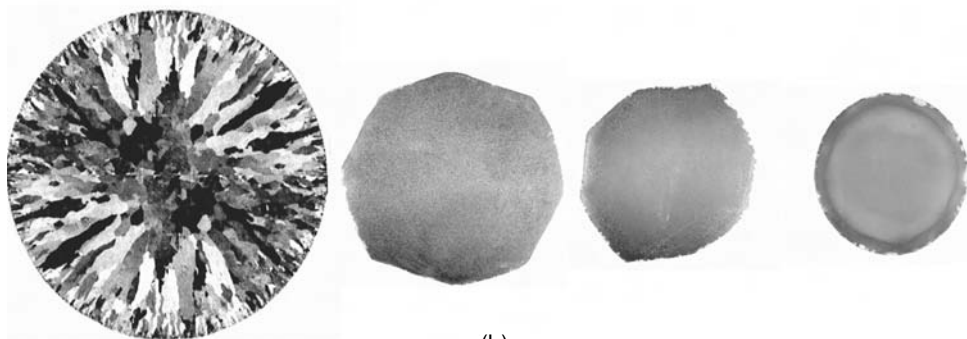
To reduce a 60 cm diameter ingot down to the required 30 cm or so as many as seven heats may be required. In the later heats a temperature below the relevant solvus (for example, 1040 °C for the γ' in Waspaloy, 1010 °C for the δ in IN718) is employed to ensure that grain growth following deformation is inhibited. Usually, a final rounding operation is performed prior to slicing the billet for the subsequent forging operations. Typically the starting grain size is around ASTM000 (1 mm+), but, by the start of the third heat, the grain size has been reduced to around ASTM0 (0.35 mm). At the end of the process the grain size is ASTM5 or 6 (40–60 μm). Other ingot-to-billet conversion processes are possible, for example, gyratory forging machine (GFM) or radial forging, the principal difference being the strain profile imparted during processing.

D Open- and closed-die forging

Having achieved a fine-grained uniform structure through the conversion process, the objective of the forgemaster is to turn the cylinder of metal into a shape close to that of the turbine disc. The principal concern at this stage is not to promote excessive grain growth. Again the production of the disc is usually a three-stage process, typically comprising (i) upsetting by open-die forging, to produce an axisymmetric ingot of reduced thickness and greater cross-sectional area, (ii) blocking by closed-die forging, to place thicker and thinner sections in the appropriate place, and (iii) finishing, again by closed-die forging, to produce the desired disc shape. In these three operations the temperature is typically kept



(a)



(b)

Fig. 4.11. (a) Cogging operation for the thermal-mechanical working of a superalloy billet. (b) Cross-sections of the worked billet after various stages of working – with the unworked billet at left. (Courtesy of Mark Roberts.)

very close to the appropriate solvus of the alloy being forged, thereby ensuring that any recrystallisation that does occur results in a finer grain structure than that achieved in the conversion stage. At the end of processing the forged discs are either quenched into a bath of oil or water, or else left to cool on the shop floor.

A heat treatment is then applied to encourage the appropriate strengthening phases (for example, γ') to precipitate in sufficient quantities to give the necessary high-temperature properties. A quenching operation, typically into oil, is employed following the heat treatment to 'freeze in' the precipitation structure. Next, the discs are machined to a pre-specified geometry which is known as the 'condition of supply'; this is usually rectilinear to allow

non-destructive evaluation (NDE) of the forging via ultrasonic testing. Finally, the rectilinear forging is machined to the final shape required for the turbine disc. These steps are obviously such that the mass of material in the final machined disc is only a small fraction of that in the turbine disc forging from which it is made. Consider, for example, an HP turbine disc for Rolls-Royce Corporation's AE3007 engine which powers the Cessna Citation X business jet. The final disc weighs about 8 kg, but it is machined from a disc of mass 45 kg.

Case study. Process modelling of forging operations

Although superalloys have been used for the turbine discs in jet engines for about 50 years, the processing is not without some challenges. The manufacturing stream is quite lengthy, involving a number of material suppliers, melters and forgemasters; at each stage there is a risk that defects will be introduced. Whilst these are usually detected by quality control procedures such as ultrasonic inspection, they nonetheless contribute to material wastage and hence to extra cost. Possible defects include (i) inadequate control of grain size and (ii) forging defects such as laps or inadequate filling of the dies.

Process modelling is one way of responding to this situation. Here, one builds mathematical models of the phenomena occurring during the manufacturing route. This should include continuum phenomena such as heat flow as well as the stresses and strains which are developed during deformation. However, it is becoming increasingly possible to simulate microstructure evolution such as recrystallisation, grain growth and texture. The emphasis is always on including as much as possible of the underlying theory, without making the simulations unwieldy, intractable or lengthy. It is important to have accurate thermophysical properties of the material, and a good estimate of the stress-strain behaviour at elevated temperatures.

There are many advantages of building a process model for the forging processes, as follows. A model, if sufficiently accurate, will have considerable predictive capability; see Figure 4.12. Models can be used to study the effect of changing process variables (such as die geometry, bite size, quenching rate) without resorting to experimental trials. They can therefore be used to optimise the processing in order to eliminate or reduce manufacturing waste. Furthermore, they generally help to reduce the number of design/make iterations required during component prototyping – this leads to savings in both time and money.

4.1.2 Processing by the powder route

The need for powder metallurgy (P/M) [12] for the production of superalloy turbine discs became apparent in the 1970s due to the development of alloys such as Rene 95, Astroloy and MERL 76 (with strength levels higher than earlier alloys such as IN718 and Waspaloy), which were found to be unamenable to processing into the cast-and-wrought form. Initially, the use of P/M product was limited to gas turbines for military purposes, for example, the Pratt & Whitney F100 engine for the F-15 Eagle fighter, which entered service in 1974 [13], but commercial applications such as General Electric's CF6-80 engine for the Boeing 747-400 and the GE90 for the Boeing 777 have emerged as experience has been accumulated.

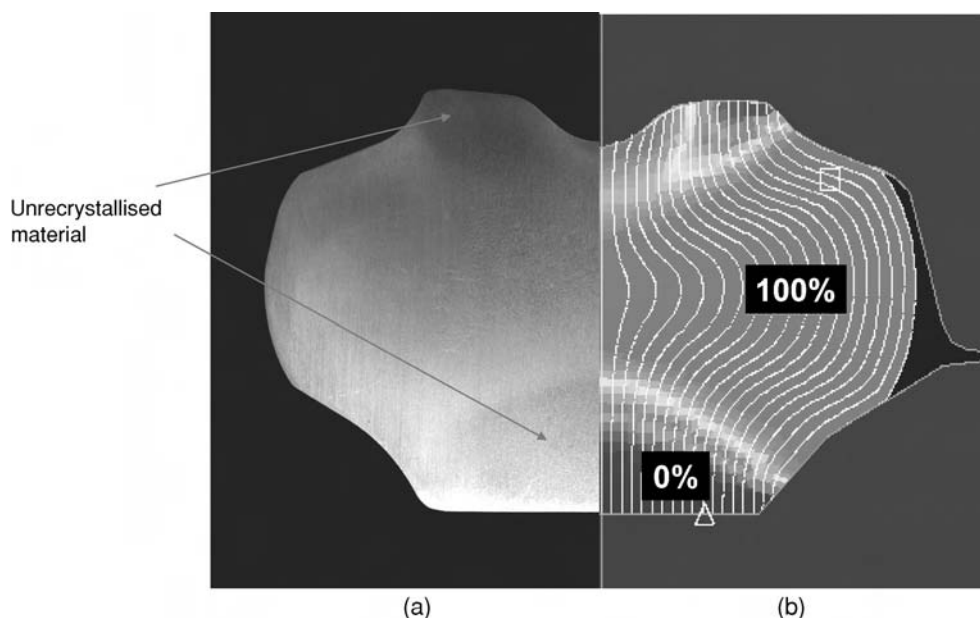


Fig. 4.12. (a) Photomicrograph of a closed-die forging in the IN718 superalloy; (b) computer simulation of the recrystallised regions in the forging, with the dead zones at top and bottom. (Courtesy of Rob Guest.)

Advantages of powder-processing include improved property uniformity due to the elimination of macrosegregation and the development of a fine grain size. The disadvantages are the extra cost associated with the production of the powder (this is offset somewhat by the number of processing steps being fewer) and, arguably, the extra complexity of the probabilistic lifing method required; see Section 4.3.

The first step in the production of P/M product is inert gas atomisation from master melts produced by VIM processing; clearly, the cleanliness of these is important since any contamination will be inherited by the powder. A typical arrangement is shown in Figure 4.13. Molten metal is poured into a tundish, which contains a carefully designed ceramic nozzle surrounded by one or more inert gas jets; usually argon is preferred. A continuous stream of gas at high pressure is delivered to the stream of molten metal arriving at the nozzle, the action of which causes its disintegration into spherical particles of diameter 30 to 300 μm . The atomisation chamber is designed to be long enough to allow solidification of the particles before they reach the outlet of the chamber. Cooling rates are typically in excess of 100 $^{\circ}\text{C/s}$, being greater for finer particles. Figure 4.14 shows a SEM micrograph of powder produced in this way. A number of variants of the process have been developed with varying degrees of success, for example, soluble-gas atomisation in which atomisation is induced by inserting a ceramic tube into the melt, the open end of which is connected to a vacuum chamber; the dissolution of a soluble gas (usually hydrogen) over the molten metal aids the atomisation process.

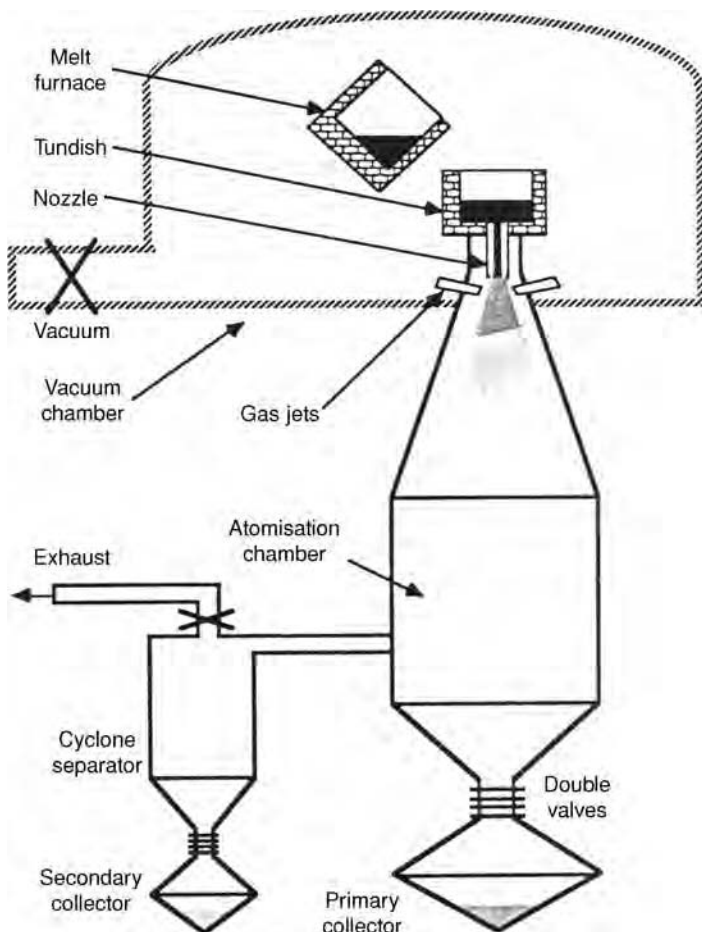


Fig. 4.13. Typical arrangement for the Ar atomisation used for the production of superalloys by powder metallurgy. (Courtesy of Tim Howson.)

Consolidation of the powder involves processing by extrusion. The fundamental challenge is to obtain good metallurgical bonding across the prior particle boundaries; to ensure bonding across oxide films which are inevitably present, the particles must be brought into contact at temperature and subjected to pressure and mechanical deformation. The powder is first packed into a carefully cleaned steel container. This is evacuated to encourage outgassing, sealed and then compacted either by hot isostatic pressing (HIP) or occasionally using closed-die forging. A very high-tonnage press with specialised tooling is then used to extrude the material to the high reduction ratios necessary for fine-grained billet; this step has been shown to disperse any non-metallic inclusions still present, thus improving the defect tolerance of the material. Some attempts have been made to eliminate the extrusion step, replacing it, for example, with a sub-solidus, hot isostatic pressing (SSHIP) process [14]; this is followed by conventional ingot-to-billet conversion of the type employed for

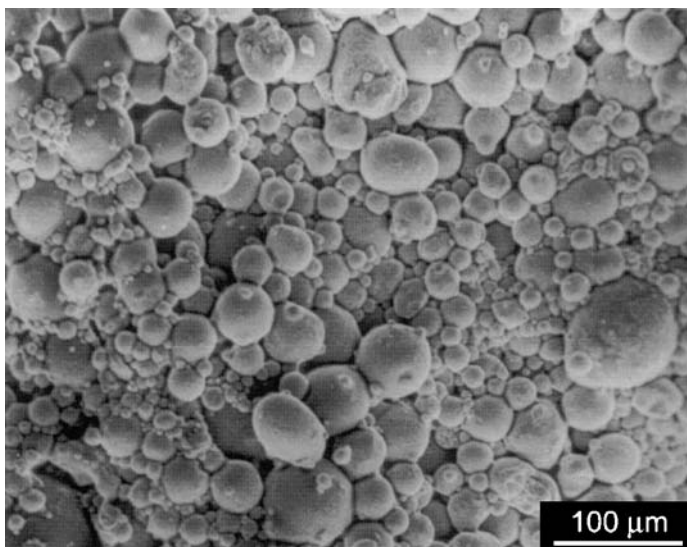


Fig. 4.14. Scanning electron micrograph of gas atomised IN718 powder.

superalloys produced by ingot metallurgy, involving thermal-mechanical working by cogging and closed-die forging.

Isothermal forging – in which the billet is deformed at the same temperature as the dies, both being held in a well-instrumented furnace – is the preferred route by which P/M superalloy product is shaped after powder consolidation. Superplastic behaviour is observed on account of the small grain size and because the deformation rates are relatively low (in the range 0.002 to 0.03/s) and stresses are high (50 to 100 MPa at 1200 °C). Several advantages then arise. Recrystallisation and grain growth can be controlled; for example, the considerable variations in microstructure arising from cold-die forging, due to excessive grain growth and die-chill, are avoided. Second, significant savings in material can be made since forgings can be produced of a geometry very close to the ‘sonic shape’ required by ultrasonic testing. Figure 4.15 illustrates possible processing sequences for the compressor discs of the GE F-101 engine, which was one of the first applications of the Rene 95 P/M superalloy [13].

When P/M processing was first applied to the production of turbine discs, the material was susceptible to the presence of defects – this was problematical in view of the need for fatigue resistance. Although the situation has improved very markedly, it is imperative to guard against the possibility of their presence. Consequently, the highest standards of handling are required, processing steps need to be strictly adhered to and quality assurance procedures must be enforced. For example, residual voids, pores and prior particle boundaries (PPBs) will be present if the steps taken to consolidate the powder are inadequate. Contamination by metallic inclusions remains a possibility, for example, by cross-contamination with a foreign alloy powder; this eventuality can be minimised by cleaning the chamber using washing runs. Of greatest concern is the possible presence of ceramic inclusions in the material,

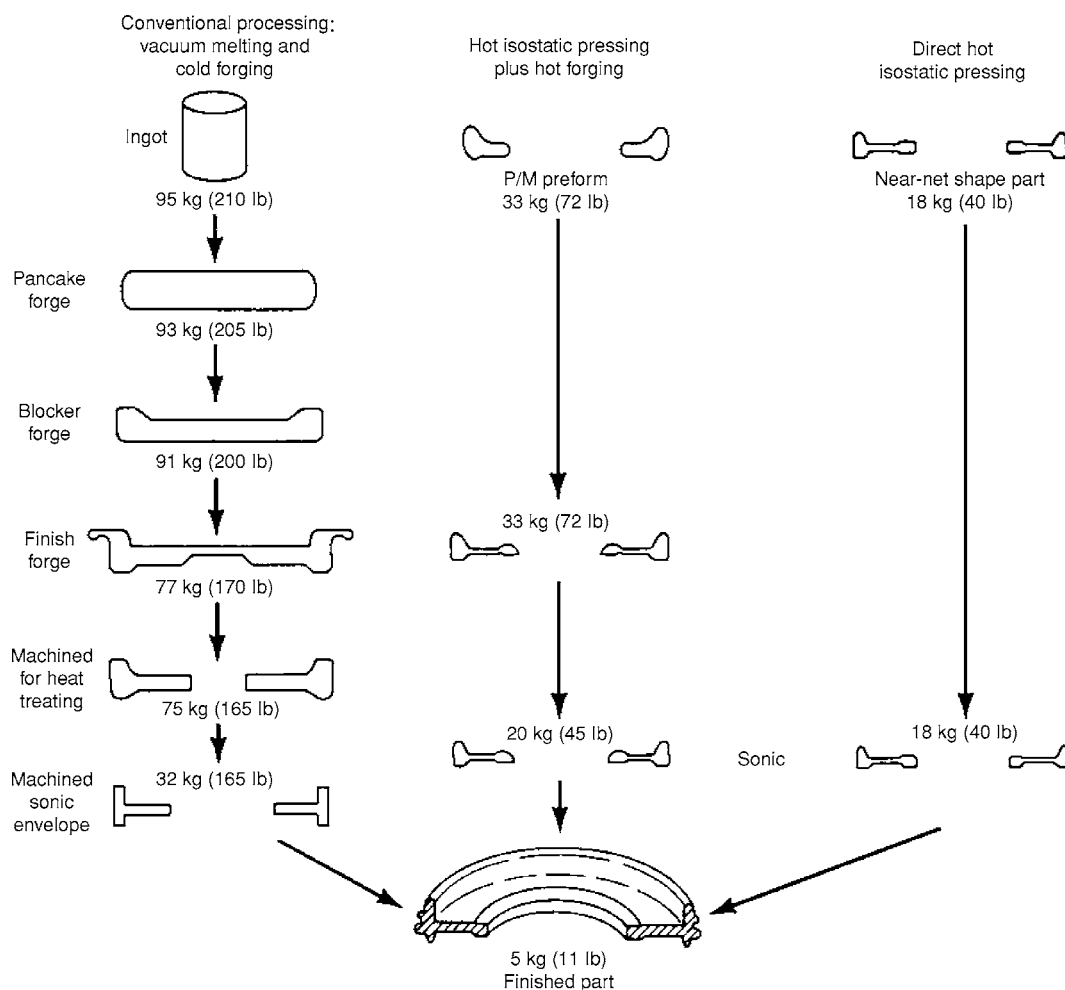


Fig. 4.15. Processing sequences for the production of F-101 compressor discs [13].

due to erosion of the melt crucible, tundish or nozzle. This is the major reason for the use of screening, which involves the passing of powder through filters of different mesh sizes; this removes the oversized particles, into which category the most damaging non-metallic inclusions fall. For most applications, the allowable size of powder is less than 230 mesh (63 μm), but, for the most stringent applications, it can be as low as 325 mesh (45 μm); obviously at such fine sizes the efficiency of the process is necessarily compromised. Blending of different powder sizes is commonplace.

The benefits of powder-processing over ingot metallurgy which produces cast-and-wrought product are now well established. For example, the low-cycle-fatigue (LCF) properties of Rene 95 at 650 °C have been compared [15] in the HIP, HIP + isothermally forged, extruded + isothermally forged and cast-and-wrought conditions; see Figure 4.16. The

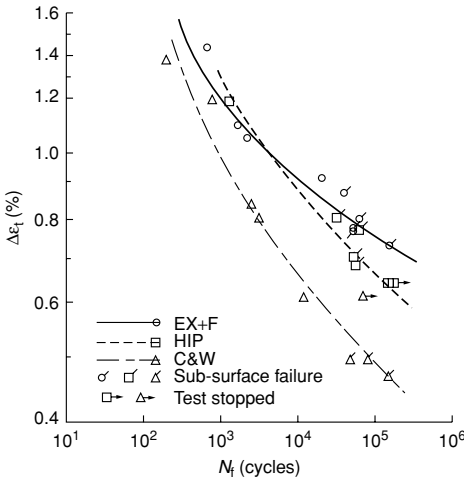


Fig. 4.16. Comparison of low-cycle-fatigue data for the Rene 95 superalloy at 650 °C, in the powder and cast-and-wrought (C&W) forms [15]; EX+F refers to extrusion and isothermal forging, and HIP is hot isostatic pressing.

best properties were displayed by extruded-and-forged material and the worst by the conventional cast-and-wrought processing, with the behaviour of the HIP material between the two; the differences were most pronounced at low total strain ranges. It was noted that, in all cases, the LCF lives were limited by the presence of ceramic inclusions – it was recommended that much further effort be expended to reduce or eliminate these defects.

4.2 Composition, microstructure and properties of turbine disc alloys

Our considerations indicate that the operating temperatures for turbine discs are considerably lower than for turbine blading, but that the stresses experienced are much greater. Under these circumstances, the properties required of a turbine disc alloy can be summarised as follows: (i) high yield stress and tensile strength to prevent yield and fracture, (ii) ductility and fracture toughness, to impart tolerance to defects, (iii) resistance to the initiation of fatigue cracking and (iv) fatigue crack propagation rates that are as low as possible. Creep resistance is also important, but traditionally it has been given less emphasis due to the lower operating temperatures and because a stress relaxation capability around notches and features of stress concentration is desirable.

What factors must be accounted for when designing a turbine disc alloy, or else when choosing one from the list of those available? Table 4.2 provides the compositions of some turbine disc alloys which are used in gas turbine engines [16]. Unsurprisingly, the chemical composition is found to be of crucial importance; however, just as importantly, if the best properties are to be attained then the microstructure must be conditioned in an optimum way. An added complexity, which does not apply to the case of the single-crystal superalloys considered in the previous chapter, is the presence of grain boundaries – these must be carefully engineered. Thus a strong appreciation of the relationship between alloy chemistry

Table 4.2. *The chemical composition of some common turbine disc alloys, in wt% [16]*

Alloy	Cr	Co	Mo	W	Nb	Al	Ti	Ta	Fe	Hf	C	B	Zr	Ni
Alloy 10	11.5	15	2.3	5.9	1.7	3.8	3.9	0.75	—	—	0.030	0.020	0.05	Bal
Astroloy	15.0	17.0	5.3	—	—	4.0	3.5	—	—	—	0.06	0.030	—	Bal
Inconel 706	16.0	—	—	—	2.9	0.2	1.8	—	40.0	—	0.03	—	—	Bal
Inconel 718	19.0	—	3.0	—	5.1	0.5	0.9	—	18.5	—	0.04	—	—	Bal
ME3	13.1	18.2	3.8	1.9	1.4	3.5	3.5	2.7	—	—	0.030	0.030	0.050	Bal
MERL-76	12.4	18.6	3.3	—	1.4	0.2	4.3	—	—	0.35	0.050	0.03	0.06	Bal
N18	11.5	15.7	6.5	0.6	—	4.35	4.35	—	—	0.45	0.015	0.015	0.03	Bal
Rene 88DT	16.0	13.0	4.0	4.0	0.7	2.1	3.7	—	—	—	0.03	0.015	0.03	Bal
Rene 95	14.0	8.0	3.5	3.5	3.5	3.5	2.5	—	—	—	0.15	0.010	0.05	Bal
Rene 104	13.1	18.2	3.8	1.9	1.4	3.5	3.5	2.7	—	—	0.030	0.030	0.050	Bal
RR1000	15.0	18.5	5.0	—	1.1	3.0	3.6	2.0	—	0.5	0.027	0.015	0.06	Bal
Udimet 500	18.0	18.5	4.0	—	—	2.9	2.9	—	—	—	0.08	0.006	0.05	Bal
Udimet 520	19.0	12.0	6.0	1.0	—	2.0	3.0	—	—	—	0.05	0.005	—	Bal
Udimet 700	15.0	17.0	5.0	—	—	4.0	3.5	—	—	—	0.06	0.030	—	Bal
Udimet 710	18.0	15.0	3.0	1.5	—	2.5	5.0	—	—	—	0.07	0.020	—	Bal
Udimet 720	17.9	14.7	3.0	1.25	—	2.5	5.0	—	—	—	0.035	0.033	0.03	Bal
Udimet 720LI	16.0	15.0	3.0	1.25	—	2.5	5.0	—	—	—	0.025	0.018	0.05	Bal
Waspaloy	19.5	13.5	4.3	—	—	1.3	3.0	—	—	—	0.08	0.006	—	Bal

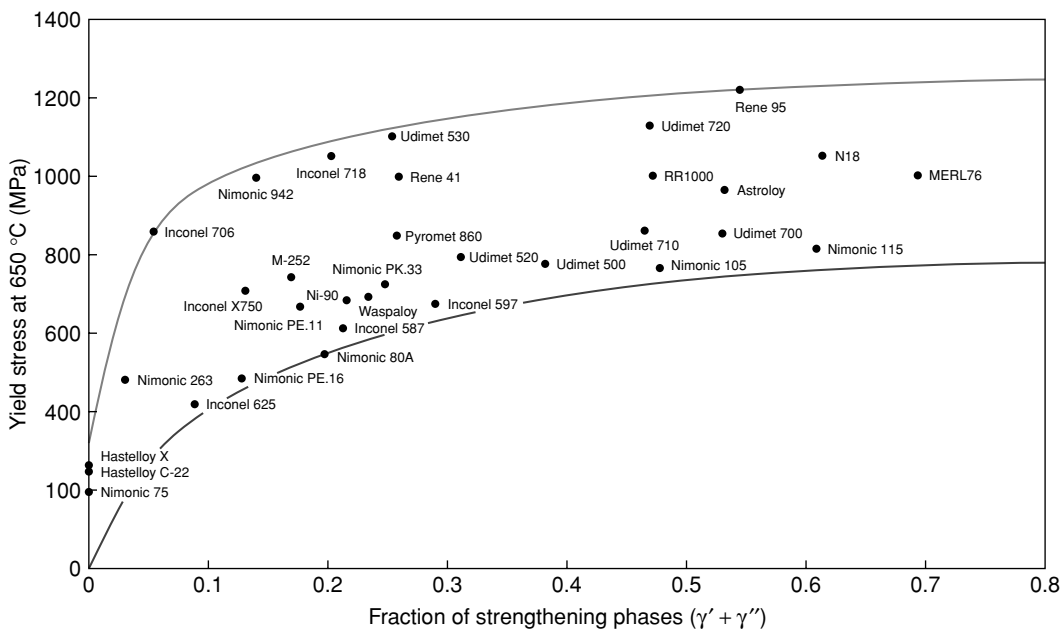


Fig. 4.17. Variation of the yield stress at 650 °C with the total fraction of the γ' and γ'' strengthening phases, for a number of common turbine disc alloys.

and microstructure is required if an optimum combination of properties is to be achieved. The experience gained so far indicates that the following three guidelines should be adhered to.

4.2.1 Guideline 1

To impart strength and fatigue resistance, the fraction of the γ' phase should be optimised by appropriate choice of the γ' -forming elements (Al, Ti and Ta) – placing it in the range 40% to 55% – and heat treatments chosen to promote a uniform distribution of γ' particles.

The yield stresses of the turbine disc alloys correlate very strongly with the proportions of the strengthening phases γ' and γ'' , and thus a first overriding concern when designing a turbine disc alloy is the choice of the concentration of elements which promote these phases, such that the desired strength levels are achieved. One can see this by using thermodynamic software such as Thermo-Calc to estimate the fractions of γ' and γ'' present in the alloys given in Table 4.1, and comparing the predictions with mechanical property data. The results, see Figure 4.17, confirm that the yield stress at 650 °C correlates very strongly with the sum of the fractions of the strengthening phases; the yield stress at ambient temperature behaves in the same way. The creep performance, for example the stress for 1000 h life at

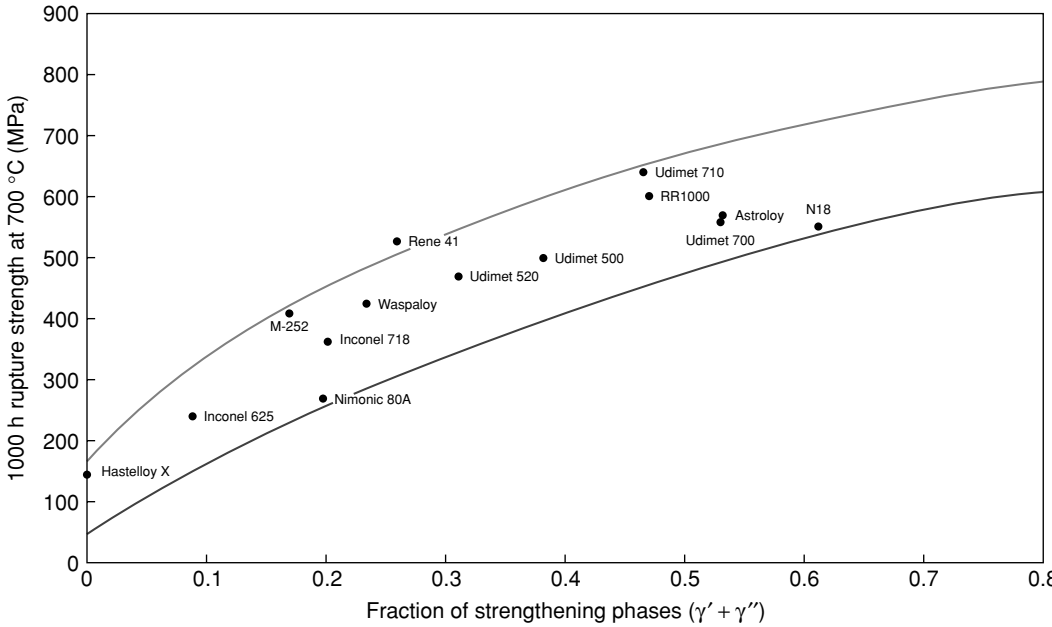


Fig. 4.18. Variation of the 1000 h creep rupture strength at 700 °C with the total fraction of the γ' and γ'' strengthening phases, for a number of common turbine disc alloys.

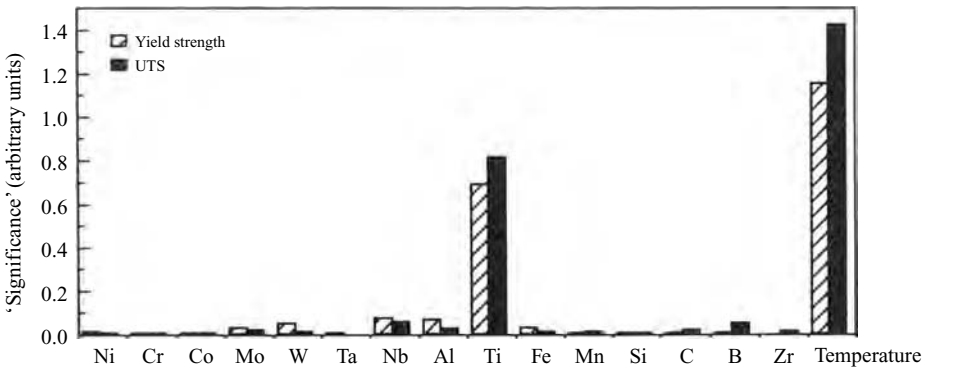


Fig. 4.19. The relative 'significance' of the elemental concentrations and temperature for a neural network analysis of the yield and ultimate tensile strength (UTS) of turbine disc superalloys [17].

700 °C, shows a similar correlation; see Figure 4.18. These findings have been confirmed by a neural network analysis [17] on a dataset containing information on the compositions and strength levels of over 200 polycrystalline superalloys; this has confirmed that the concentrations of Al, Ti and Nb are the most influential in improving the yield properties; see Figure 4.19. Interestingly, when the neural network was interrogated after being trained, it predicted that higher Ti/Al ratios are important in conferring strength (see Figure 4.20); this is consistent with the hardening theories of Chapter 2, which emphasise the importance of the APB energy, which increases as the Ti/Al ratio is raised [18]. Cobalt was found to

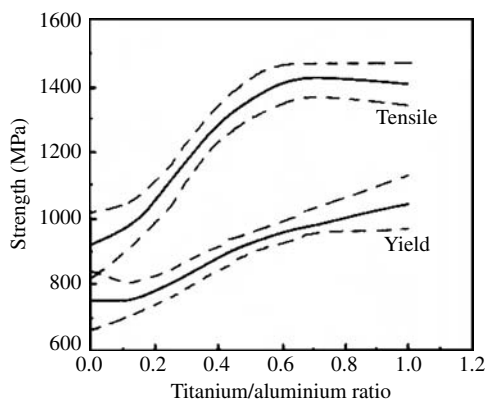


Fig. 4.20. Predicted sensitivity of the room temperature yield and tensile (UTS) strength to variations in the Ti/Al ratio, using a neural network analysis [17].

have negligible effect on the yield properties, but increasing proportions of molybdenum were predicted to be beneficial – using Astroloy as a baseline, an increase in concentration from zero to 20 wt%Mo was predicted to improve the yield stress by about 500 MPa. This information about the beneficial effect of Mo was used to help optimise the composition of the RR1000 superalloy; see Table 4.1. The importance of a high Ti/Al ratio in conferring strength can be further demonstrated by considering the data; if one plots the yield stress against the root of the sum of the strengthening phases, a reasonably straight line is found, consistent with the hardening theories of Chapter 2 – one can then determine the extent to which any one alloy lies above or below this ‘mean’, or expected, behaviour. The results are given in Figure 4.21. One can see that the alloys with high Ti/Al ratio are the ones which possess the highest strength, consistent with the neural network analysis of [17].

In practice, the yield stress of a turbine disc alloy has been found to depend critically on the heat treatment applied to it – particularly the solutioning treatment, which is carried out in the vicinity of the γ' solvus. This is because, as indicated by the theoretical analysis given in Chapter 2, the size of the γ' particles is as crucial as their volume fraction in determining the resistance to dislocation motion. For most turbine disc alloys, the γ' solvus lies between 1050 °C and 1200 °C, the exact value being dependent upon the chemical composition. If the solutioning heat-treatment temperature is sub-solvus, the undissolved, or *primary*, γ' precipitates have a grain-pinning effect which restricts γ -grain growth; see Figure 4.22. Super-solvus heat treatment eliminates the primary γ' and the γ grains are correspondingly coarser. It follows that the γ' phase distribution is, in practice, rather complex; see Figure 4.23, which is a schematic illustration of the microstructure of a turbine disc alloy [19]. In addition to the primary γ' , intragranular secondary and tertiary γ' particles are present within the γ grains, typically at particle diameters of ~ 100 nm and ~ 50 nm, respectively. This bimodal distribution occurs because of the interplay between the kinetics of γ' particle nucleation, growth and coarsening, which occurs during isothermal solutioning and ageing heat treatments; the cooling rate is also important since the equilibrium fraction of γ' is rarely attained during these, so that further γ' nucleation and growth occurs during cooling. During the ageing heat treatment, which is typically carried out at around 700 °C for several hours, an age-hardening response is observed, as in many precipitation-hardened

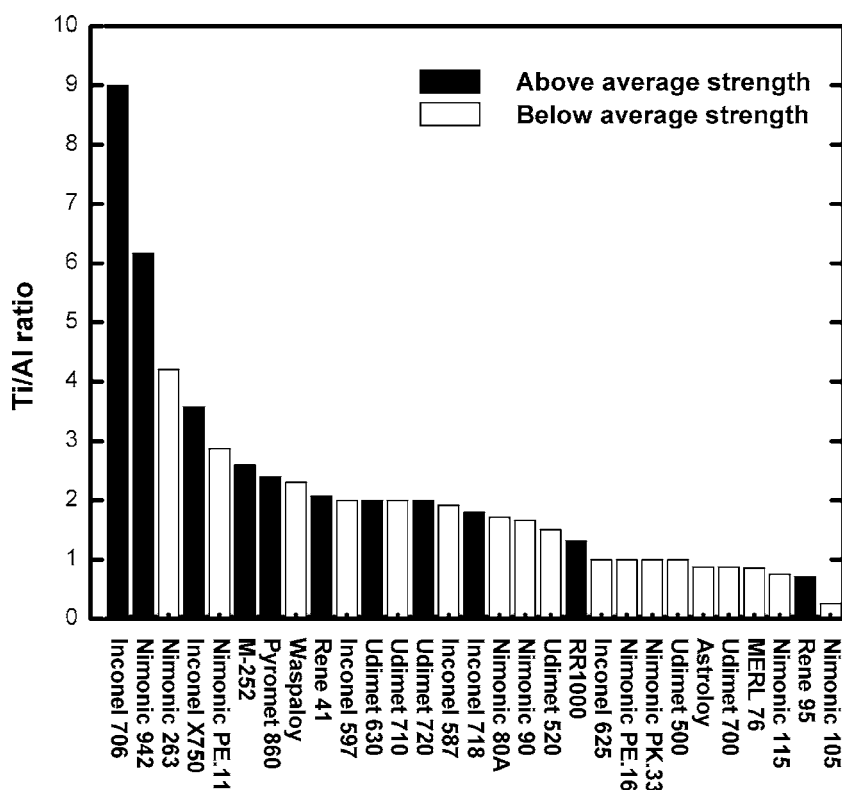


Fig. 4.21. Variation of the Ti/Al ratio in a number of polycrystalline superalloys, ranked in order of strength consistent with a deviation away from a perfect correlation of yield stress and the square root of the fraction of strengthening phases.

systems. The data in Figure 4.24 for the yield stress and UTS of the Udimet 720Li superalloy illustrate this; in this case, the length of the heat treatment is optimal at about 30 h [19]. When the γ' particle distributions giving rise to this behaviour are examined, one finds that the size of the secondary γ' particles is unaltered by the heat treatment; see Figure 4.25. On the other hand, the tertiary γ' coarsens considerably, and this proves unambiguously that it controls the strength level achieved. Clearly, the heat treatment must be chosen carefully if optimum properties are to be attained. On the basis of these observations, one can argue that, for absolute static strength, higher solutioning temperatures are to be preferred, since the proportion of primary γ' is then reduced, so that a greater proportion of the available γ' is then placed in the range 50 nm to 100 nm where it has the greatest effect.

The situation, however, is complicated because the fatigue properties must also be considered. The distribution of the γ' precipitates has been found to have a significant influence on the dwell-time fatigue crack propagation (FCP) rate; for example, an improvement in FCP rate in the Paris regime of about an order of magnitude has been reported in ref. [20] when heat treatment was super-solvus rather than sub-solvus, this microstructural effect being more potent than minor changes in chemistry involving Nb/Ta ratio and Co content

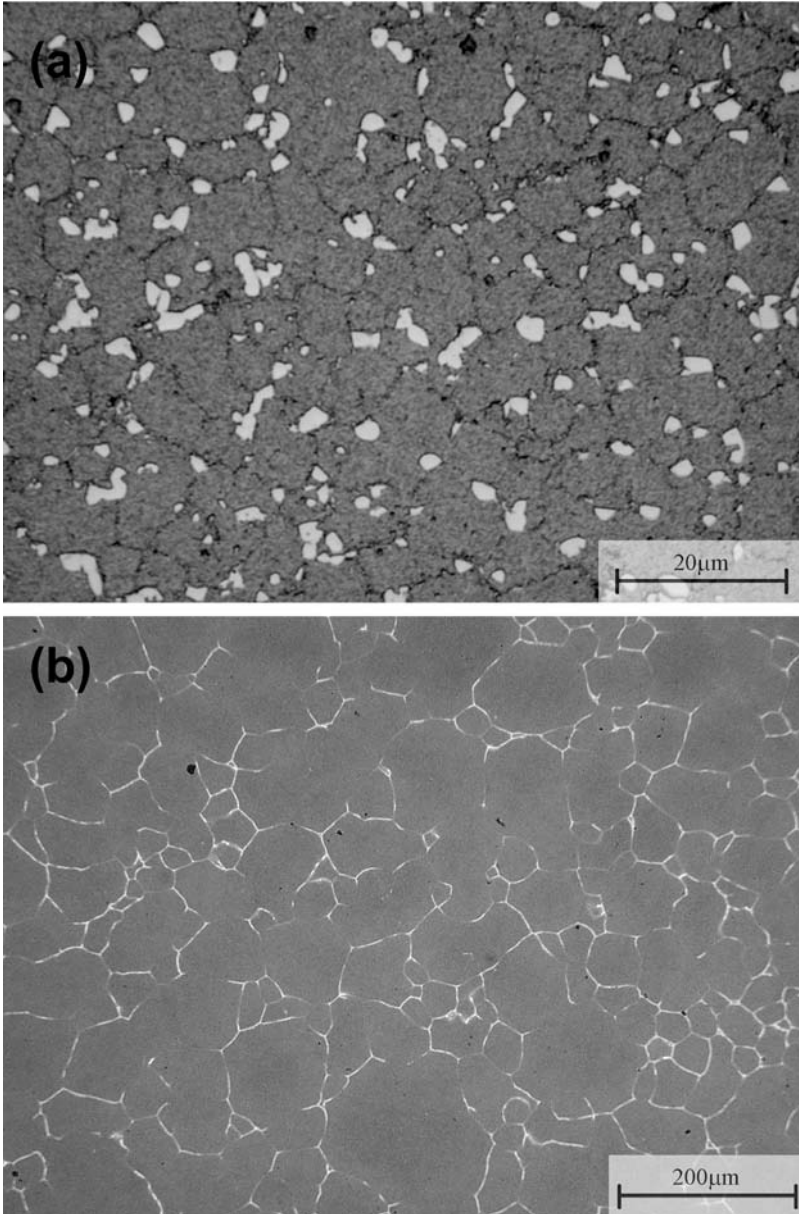


Fig. 4.22. Optical micrographs of RR1000 superalloy: (a) sub-solvus heat-treated for 4 h at 1130 °C; (b) super-solvus heat-treated for 4 h at 1170 °C. The solvus temperature in this alloy is about 1150 °C. Note the grain-pinning effect, which occurs when the heat-treatment temperature is sub-solvus. (Courtesy of Rob Mitchell.)

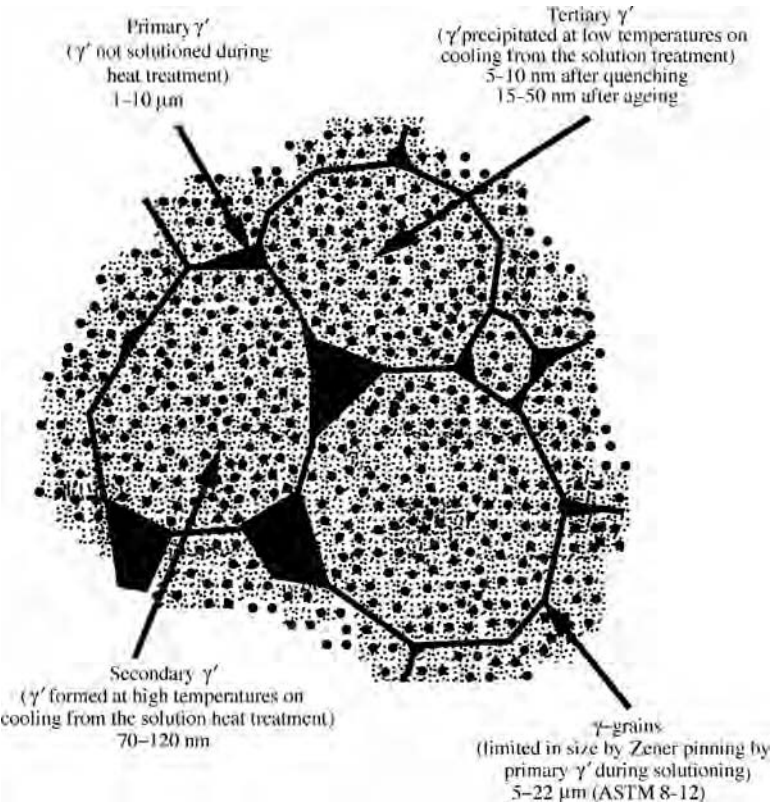


Fig. 4.23. Schematic illustration of the distribution of the γ' phase in a turbine disc alloy [19].

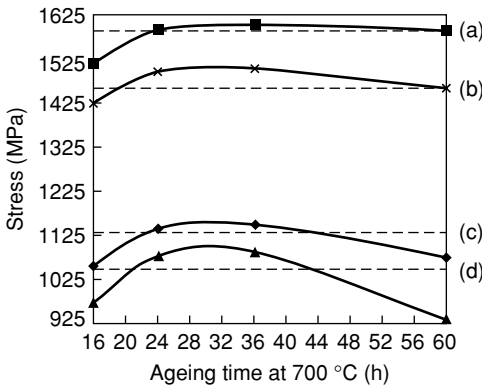


Fig. 4.24. Hardening response of the Udimet 720Li superalloy due to heat treatment at 700 °C [19]. (a) Room temperature, UTS; (b) room temperature, 0.2% proof stress; (c) UTS at 600 °C; (d) 0.2% proof stress at 600 °C.

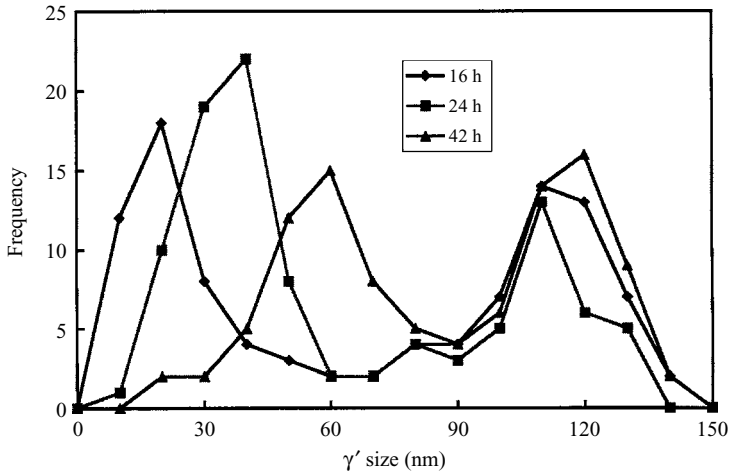


Fig. 4.25. Characterisation of the bimodal distribution of γ' in the Udimet 720Li superalloy and its response to ageing at 700 °C [19].

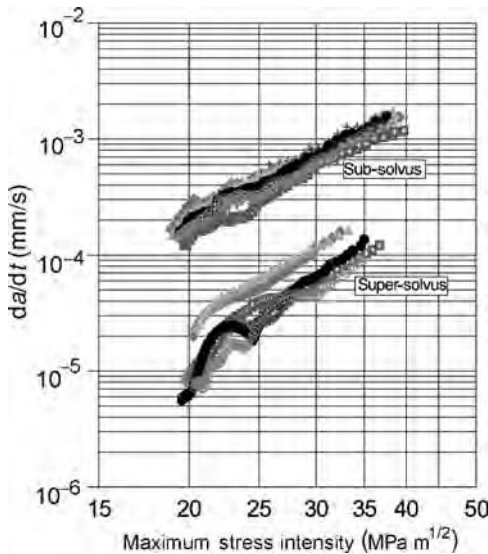


Fig. 4.26. Dwell fatigue crack propagation data for seven P/M superalloys at 704 °C and a 90 s hold, with $R = 0.1$ [20]. The base composition was Ni–11Cr–2.6Mo–3.8Ti–3.9Al–5.7W, with 0.04, 0.03 and 0.1 of the trace elements C, B and Zr; Nb, Ta and Co were varied between 0 and 1, 1 and 2 and 15 and 19, respectively. All figures in wt%. Super-solvus processing clearly has a greater influence than modifications in the composition of the alloy.

(see Figure 4.26). When super-solvus processing was employed, the cooling rate was also found to be important at constant γ -grain size; the FCP rate increased by about a factor of 4 at a K_{max} of 25 MPa m^{1/2} when air cooling rather than furnace cooling was employed – because the γ' particle size decreased from 250 nm to 170 nm; see Figure 4.27. Combinations of the original super-solvus heat treatment followed by subsequent sub-solvus heat treatment were shown to be very detrimental, with a further increase in FCP rate of at least 3 – this can be attributed to the dissolution of the larger γ' particles and the reprecipitation of smaller ones upon cooling. From these observations it seems that the γ' particle size

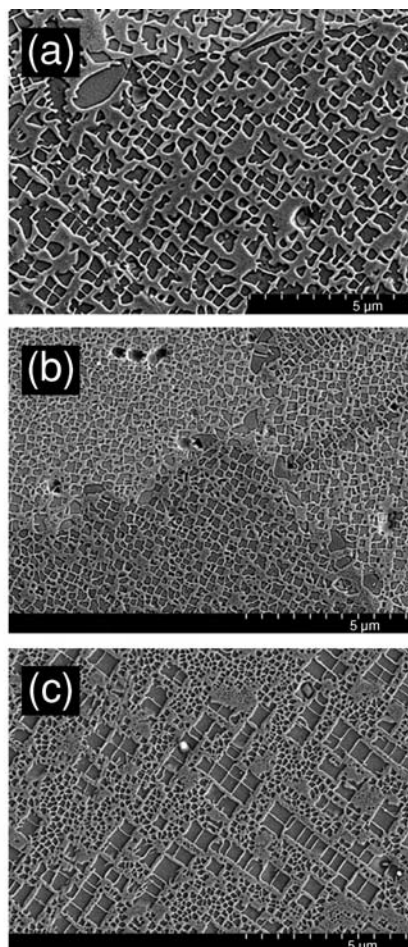
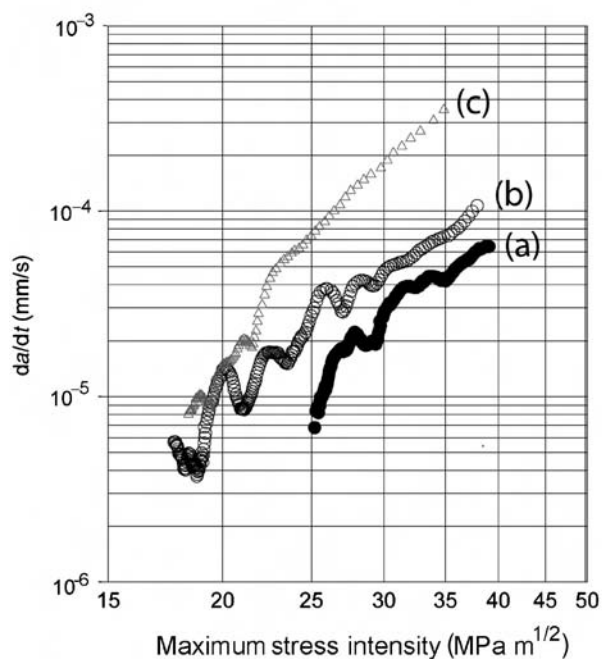


Fig. 4.27. Dwell fatigue crack propagation data at 704 °C (90 s hold and $R = 0.1$) for an experimental P/M superalloy, heat-treated to give different γ' sizes [20]: super-solvus solutioning, followed by (a) furnace cooling, no resolution (b) air cooling, no resolution and (c) air cooling, sub-solvus resolution at 1171 °C.

distribution for optimum FCP resistance is different from that required for optimum static strength. From a practical standpoint, however, it is not usually possible to manipulate the γ' particle distribution independently of the size of the γ grains, and this represents a dilemma when designing the heat treatment for a turbine disc superalloy. Tensile strength and low-cycle-fatigue (LCF) life (controlled largely by fatigue initiation) are best when the grain size is small, but the creep properties and fatigue crack propagation rates are not optimum; these are better at large grain sizes. This is illustrated in Figure 4.28, which is adapted from ref. [21]. For a balanced portfolio of properties, a compromise needs to be struck between the need for absolute static strength and resistance to cyclic fatigue crack propagation. This challenge is discussed further in the following section.

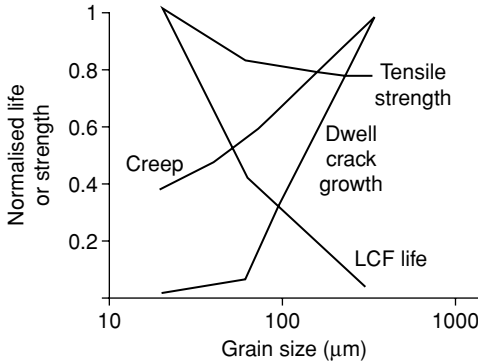


Fig. 4.28. Schematic illustration of the important properties of turbine disc alloys and their dependence upon grain size. After ref. [21].

4.2.2 Guideline 2

The grain size should be chosen for the desired combination of yield strength, resistance to fatigue crack initiation (both of which scale inversely with grain size), creep strength and resistance to fatigue crack growth (which scale directly with it). A size in the range 30 μm to 50 μm is commonly chosen.

The necessity to control the grain size has been demonstrated by a number of researchers, but a consideration of the careful work of Bain *et al.* [22] is warranted because all of the important properties – yield, fatigue and creep – were measured and reported. The Udimet 720 superalloy, processed using VIM/VAR and then combinations of hot-die or isothermal forging and sub- or super-solvus heat treatments, was considered. Identical double ageing treatments of 760 °C/8 h/air cool and 650 °C/24 h/air cool were performed in all cases. In this way, uniform grain sizes of 19 μm , 90 μm , 127 μm and 360 μm (corresponding to ASTM grain sizes of 8.5, 4, 3 and 0, respectively) were produced; unfortunately, one cannot rule out possible differences in the distributions of secondary and tertiary γ' , although the micrographs presented [22] indicate that no primary γ' was present other than in the 19 μm material. The testing of tensile specimens indicated that the finest-grained material exhibited a room temperature yield stress (YS) of about 1200 MPa, approximately 200 MPa greater than the coarsest-grained material – a figure which is approximately equal to the difference in the YS reported for Udimet 720Li in the peak-aged and over-aged conditions; see Figure 4.29 (note that the Udimet 720 and 720Li superalloys have marginally differing chemistries (see Table 4.1)). The improvement reported in the ultimate tensile strength (UTS) was of a similar magnitude. Strain-controlled LCF tests were also carried out on smooth, polished testpieces with $R = 0$ and a frequency of 0.33 Hz at a total strain range of 0.80%. The LCF results reflect the improvements in YS won by grain refinement; the LCF life was approximately one order of magnitude worse at the largest grain size of 360 μm than at 19 μm . These considerations indicate that for polycrystalline superalloys, for the best static yield and LCF properties, fine grain sizes are to be preferred. This grain size dependence of the LCF performance must be added to the intrinsic LCF strength due the

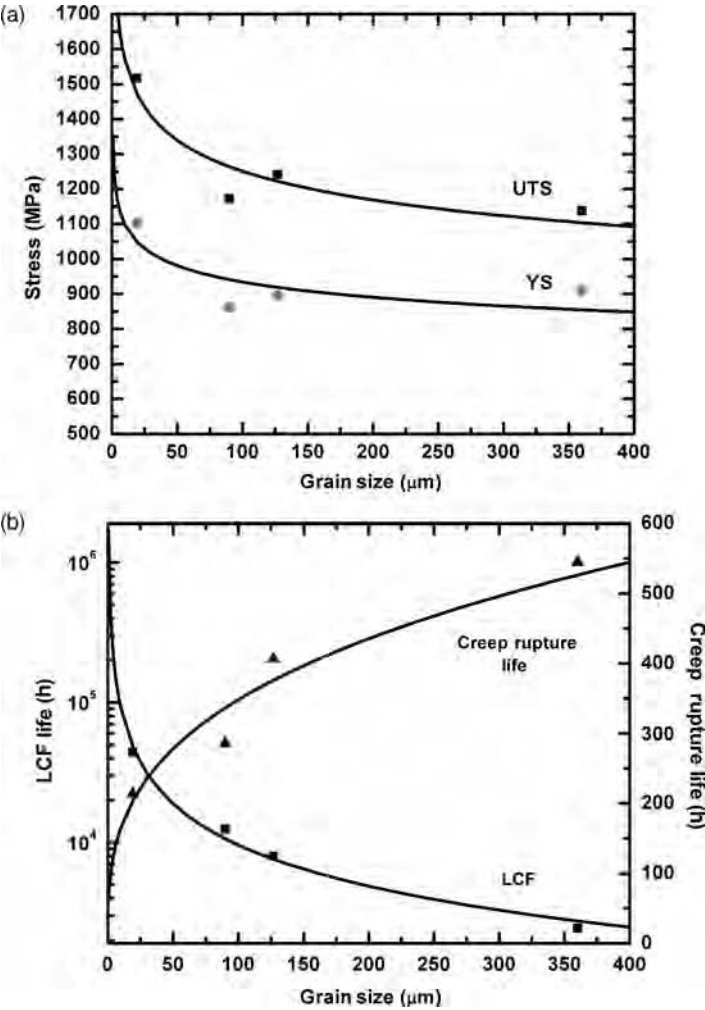


Fig. 4.29. Data for the mechanical properties of the Udimet 720 superalloy, as a function of the grain size developed by varying combinations of hot-die or isothermal forging and sub- or super-solvus heat treatments: (a) yield stress (YS) and ultimate tensile strength (UTS), (b) creep rupture life at 700 °C/690 MPa and low-cycle-fatigue (LCF) life at 650 °C at a total strain range of 0.8% with $R = 0$. Data from ref. [22].

alloy composition [23,24], which correlates strongly with the γ' fraction and thus also the yield stress; see Figure 4.30.

The influence of the grain size is very different when one considers the rates of fatigue crack growth and the creep rupture performance. Bain *et al.* [22] performed constant load creep tests at 700 °C and 690 MPa; for the largest grain size of 360 μm , the rupture life approached 600 h under these conditions, but at 19 μm it decreased to about 200 h. Similar observations have been made [25] in the NR3 superalloy heat treated at super- and sub-solvus temperatures to give coarse- and fine-grain structures, respectively (see Figure 4.31);

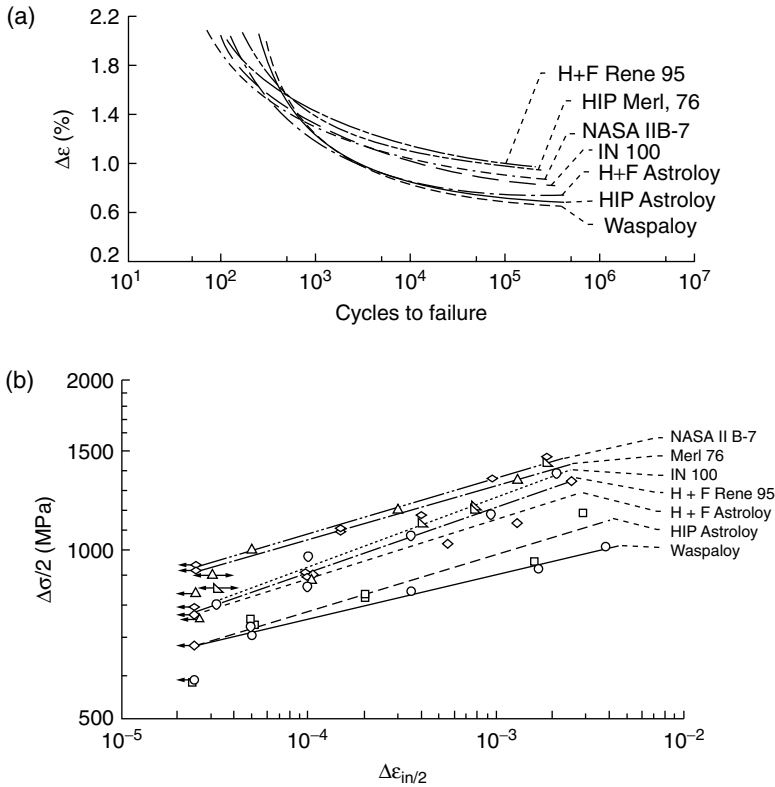


Fig. 4.30. Strain-controlled low-cycle-fatigue data ($R = -1$, 0.33 Hz) at 650 °C for various turbine disc alloys: (a) plot of total strain range vs. number of cycles to failure [23], and (b) stabilised cyclic stress-strain curves, on which the cyclic stress is plotted against the inelastic (plastic) strain range [24]. Note the poorer performance of alloys such as Waspaloy, which have a lower γ' fraction and therefore a smaller yield stress.

the minimum creep rate differed by about an order of magnitude with grain size when testing was carried out at 750 °C and 300 MPa, being better for the coarse-grain size, although the influence was less pronounced at lower temperatures and higher stresses. Thus larger grain sizes are beneficial for creep performance. The same is found to be true of the fatigue crack propagation rate; this is important if the damage-tolerant method for life assessment is being used (see Section 4.3) since its magnitude controls the fatigue life which can be declared. Fatigue cyclic crack growth rates were measured by Bain *et al.* [22] in air at 650 °C at an R ratio of 0.05, at a frequency of 3.3 Hz and with a 5 min dwell at the maximum load. A second set of tests was carried out at 425 °C without the dwell, but under otherwise identical conditions. Particularly at the higher temperature, the dependence of crack growth rate on grain size was considerable (see Figure 4.32), being about two orders of magnitude smaller for the larger grain size; consistent with the promotion of a tortuous crack path, this was attributed to rougher fracture surfaces which were largely intergranular in nature. At 425 °C, the dependence on grain size was less pronounced, but the

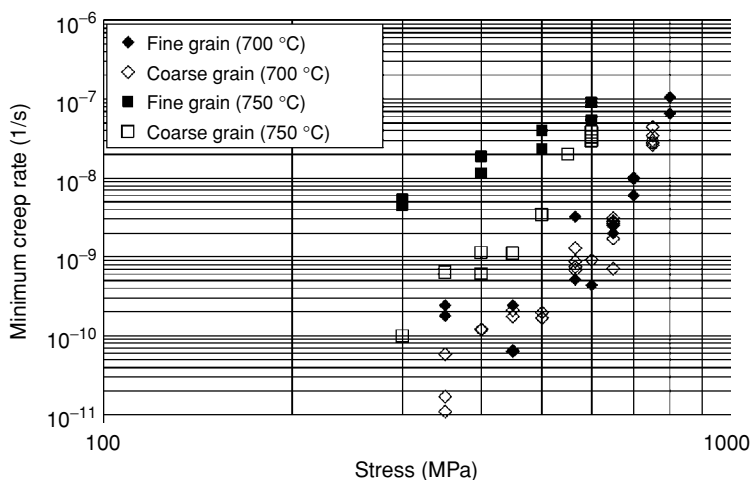


Fig. 4.31. Constant load creep data for the NR3 P/M superalloy, at temperatures of 700 °C and 750 °C in both the fine- and coarse-grained conditions [25].

smaller grain size was again found to perform better, particularly in the threshold regime at low ΔK ; the fracture surfaces were transgranular and faceted. However, at both temperatures, the Paris exponents in stage II growth were not heavily influenced by grain size. These observations are consistent with the mechanisms of fatigue crack growth which are agreed upon [23,26]: at low temperatures, high strain rates and low ΔK , there is a strong tendency for crystallographic cracking on $\{111\}$ planes due to the ‘planar slip’, which is then prevalent due to the limited number of slip systems that are activated. At higher temperatures and larger ΔK , which increases the size of the reversible plastic zone, slip is more homogeneous and ‘wavy’. Li, Ashbaugh and Rosenberger [27] have characterised these effects by examining the fracture surfaces of IN100 under LCF conditions using high-resolution SEM microscopy; see Figure 4.33. At low temperatures, fatigue crack growth is found to occur parallel to slip band traces and thus along $\{111\}$ planes (see Figure 4.33(c)); but, as the temperature increases, $\{001\}$ facets become increasingly common, as evidenced by the cuboidal morphology of the γ' precipitates on the fracture surface. Related work on IN100 [28] has demonstrated a considerable influence of environment on the fatigue crack growth rate – at 620 °C with $R = 0.15$ and a 30 s dwell at the peak load, a decrease in the vacuum from 1 to 10^{-8} atm produced an order of magnitude decrease in the crack growth rate. When incipient fatigue cracks – after the initiation period – have dimensions on the scale of the grain size, their crack growth rates are several orders of magnitude larger than predicted from the linear elastic fracture mechanics, Paris Law regime [29–31]; see Figure 4.34. Such ‘short-crack’ growth, which can occur below the fatigue crack threshold for long cracks, is of importance for turbine disc applications; this is because of the necessity for an adequate lifing policy – for example, the life-to-first-crack method described in Section 4.3 – which considers a turbine disc to have ‘failed’ before the crack reaches a length consistent with the K_{Ic} criterion. These effects are not accounted for in the traditional damage-tolerant approach.

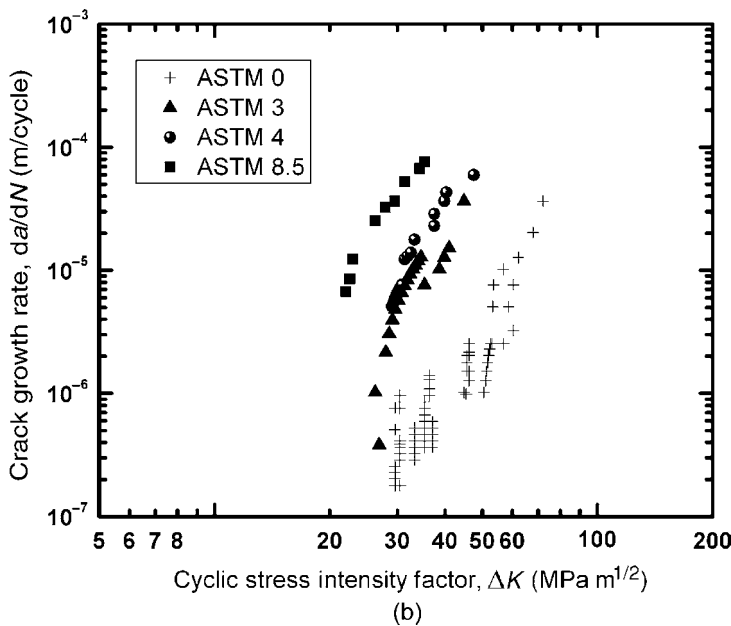
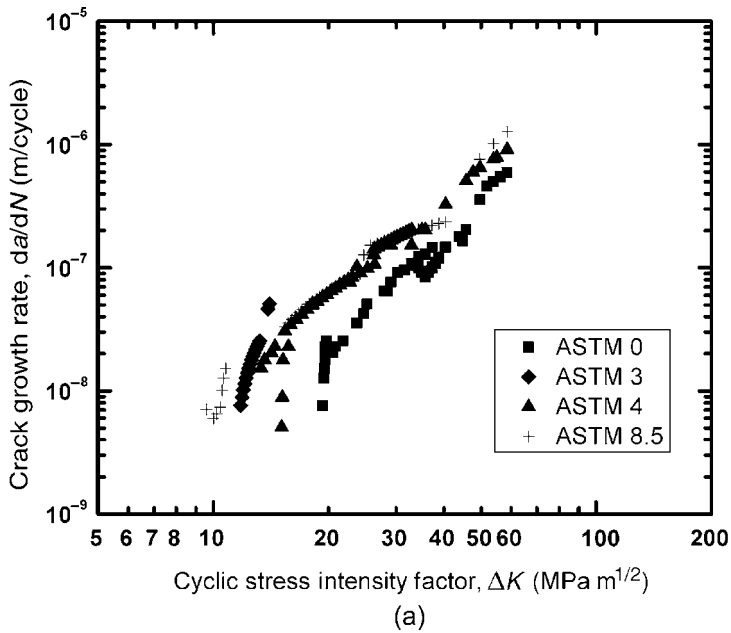


Fig. 4.32. Dwell fatigue crack propagation data for the Udimet 720 superalloy, as a function of the grain size developed by varying combinations of hot-die or isothermal forging and sub- or super-solvus heat treatments [22]: (a) 650 °C, with a 5 min dwell at peak load and (b) 425 °C. In both cases, $R = 0.05$.

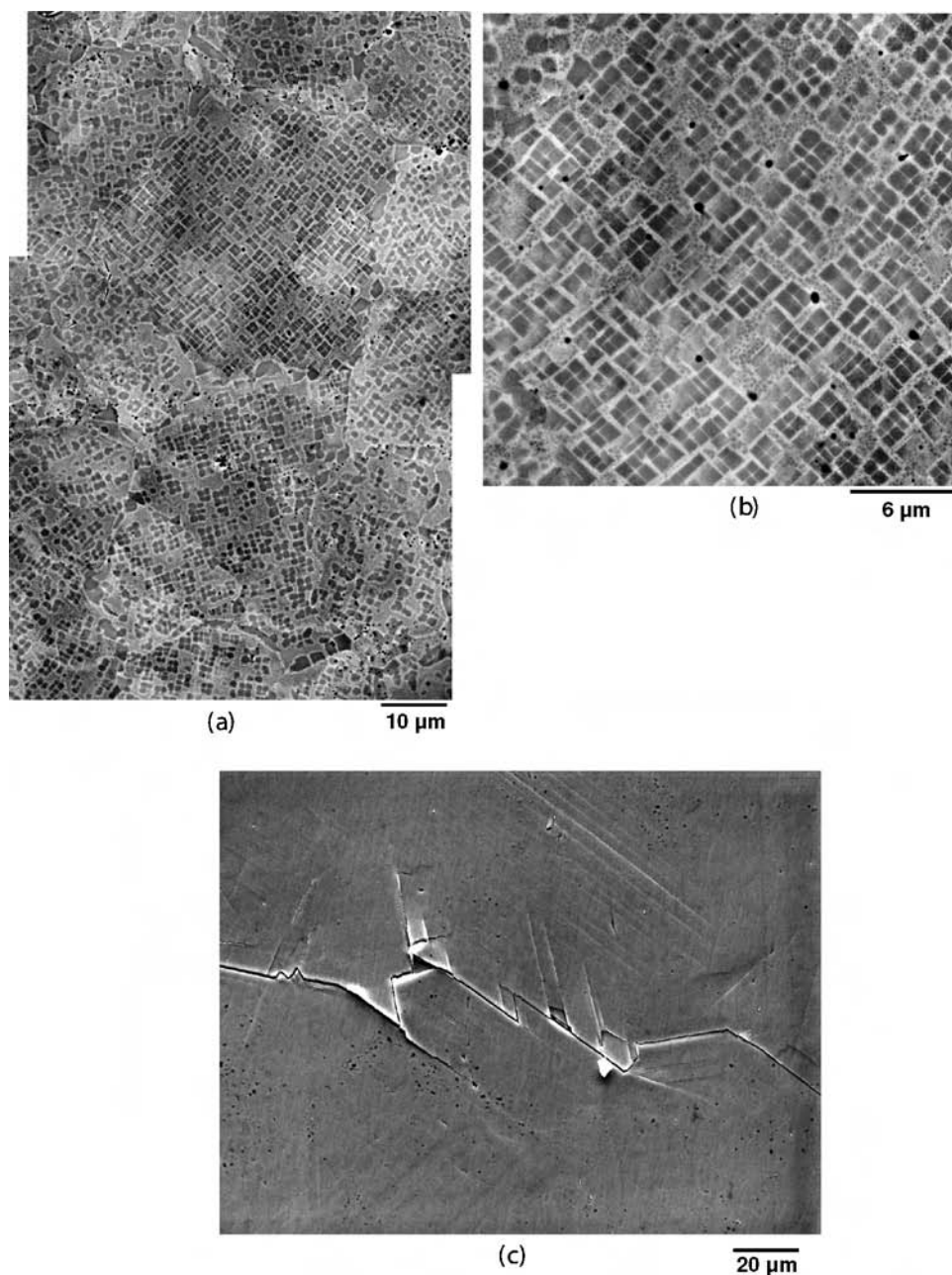


Fig. 4.33. Fracture surfaces for the IN100 P/M superalloy tested in low-cycle fatigue at 0.5 Hz with $R = 0.05$ [27]. (a), (b) SEM images of the $\{001\}$ fracture surfaces typical of failure at 538 °C, showing the presence of cuboidal facets belonging to the γ' phase. (c) $\{111\}$ fracture surface typical of fatigue crack growth at room temperature – note that the fracture path is parallel to the traces of the slip lines in the top right of the image.

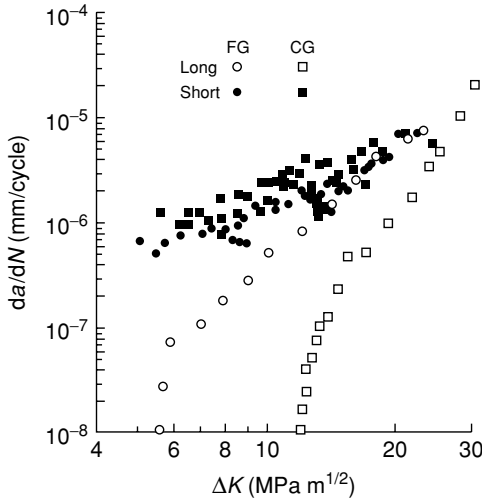


Fig. 4.34. Room-temperature fatigue crack propagation data for Astroloy at $R = 0.1$ and 40 Hz [29], for both long and short cracks; FG and CG refer to fine-grained and coarse-grained material, respectively. Note that the fatigue crack growth occurs more rapidly for short than for long cracks; also, when cracks are short, the effect of grain size is much reduced.

Because of the need for a fine grain size in the bore regions and a coarser one at the rim, so-called ‘dual heat treatment’ processes have been developed (for example, see ref. [32]). Here, a conventional furnace is first used to heat-treat the disc to a grain size appropriate for the properties required for the bore – at a temperature which is usually sub-solvus. Then, a second more sophisticated furnace is employed in which a sharp temperature gradient exists between the bore and the rim, the latter experiencing temperatures which are super-solvus; see Figure 4.35. For this purpose, cooling gas is directed at the bore regions throughout the heat treatment step, and insulating materials are placed in the furnace to promote the required temperature gradients. The use of thermocoupling and controller systems has been shown to be necessary to achieve the levels of temperature control which are required. Although this technology has yet to be used in production, trials have been carried out on the Rene 104 powder superalloy [32]; the ASTM grain size was shown to vary from 6 at the rim to about 11 at the bore, over a transition zone of about 50 mm. This approach is considered by the inventors to be advantageous for the processing of turbine discs for military applications.

4.2.3 Guideline 3

When added in small quantities, grain-boundary elements such as boron and carbon are beneficial, particularly to the creep and low-cycle-fatigue resistances.

Grain-boundary elements such as boron and carbon are beneficial when added in small quantities, since they segregate to the γ/γ interfaces and increase the work of cohesion. Improvements in the creep rupture strength, creep ductility and low-cycle-fatigue behaviour then result. However, when added in excessive quantities, the precipitation of borides and carbides is promoted; it seems likely that at these levels any further additions of the

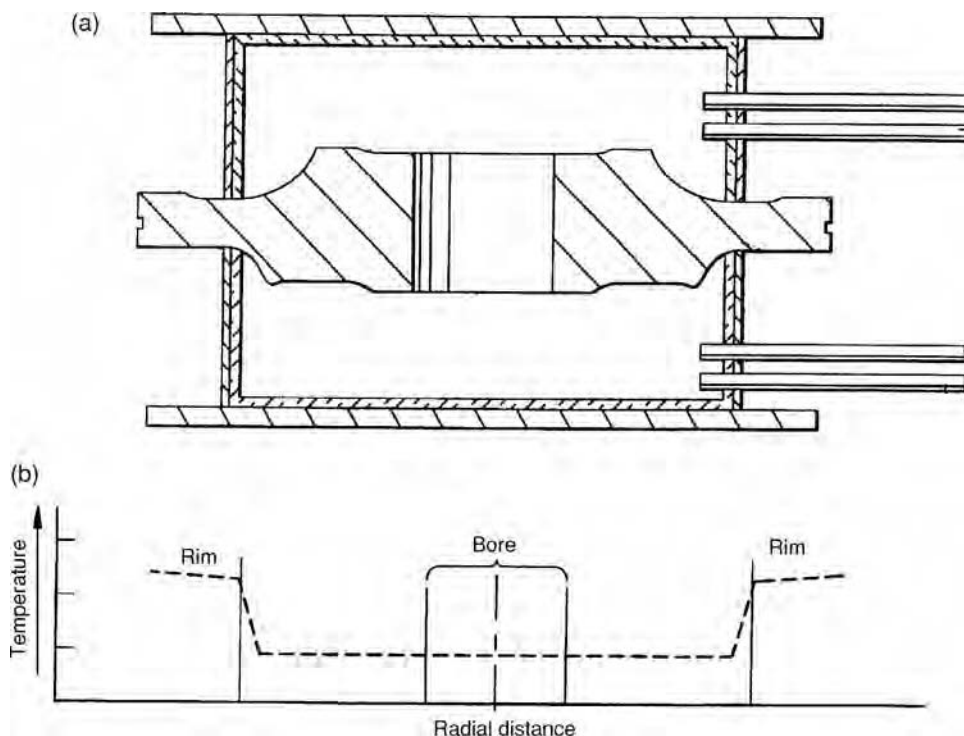


Fig. 4.35. Illustration of the arrangement for the dual heat-treatment process [32], in which a turbine disc forging is subjected to a temperature field which is deliberately engineered such that it varies from the bore to rim regions.

grain-boundary elements are ineffective. The optimum concentrations are approximately 0.03 wt%B and 0.025 wt%C; see Table 4.1. At these levels, it can be demonstrated that the γ -grain boundaries are covered with just a few monolayers of these grain-boundary elements. Other elements are often quoted as being effective at strengthening the grain boundaries. For instance, there is some evidence that zirconium, as a grain-boundary segregant, plays a role as a 'getter' of deleterious 'tramp' elements such as sulphur and phosphorus. However, it should be emphasised at this stage that the precise role played by the grain-boundary elements is controversial; more research is required.

To support these statements, consider the results reported by Garosshen *et al.* [33], who studied the mechanical properties of an experimental P/M alloy of base composition Ni-19Co-12.5Cr-5Al-4.4Ti-3.3Mo (wt%). When doped with 0.02 wt%B, 0.05 wt%Zr and 0.003 wt%C, the creep rupture life at 732 °C and 655 MPa improved remarkably from less than 0.1 h to 69 h; however, further additions of carbon to 0.06 wt% were ineffective, and additions of boron beyond 0.02 wt% reduced the rupture life, albeit only by a few hours. Boron additions over the solubility limit were shown to result in the intergranular precipitation of the M_3B_2 compound and caused no further improvement in mechanical properties, so it can be concluded that the strengthening effect of boron is not due to the formation of borides; furthermore, the additions of carbon, boron and zirconium had no effect on

Table 4.3. Alloy compositions

Alloy	Target alloy compositions (wt%)										
	C	Cr	Co	Mo	Ti	Al	W	Zr	B	Hf	Ni
Alloy 1 (baseline Udimet 720)	0.025	16.0	14.75	3.0	5.0	2.5	1.25	0.035	0.02	—	Bal
Alloy 2	0.025	16.0	14.75	3.0	5.0	2.5	1.25	0.035	0.02	—	Bal
Alloy 3	0.025	16.0	14.75	3.0	5.0	2.5	1.25	0.035	0.03	0.75	Bal
Alloy 4	0.025	16.0	14.75	3.0	5.0	2.5	1.25	0.070	0.03	—	Bal
Alloy 5	0.025	16.0	14.75	3.0	5.0	2.5	1.25	0.035	0.04	—	Bal

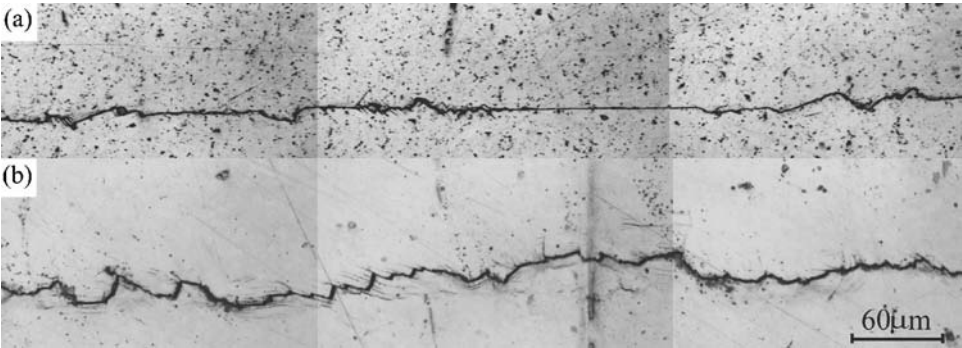


Fig. 4.36. Fatigue crack propagation paths in the IN718 superalloy doped with (a) 12 ppm B and (b) 29 ppm B [34]. Testing at room temperature in air, with $R = 0.05$.

the morphology of the γ' phase. Moreover, the fatigue crack propagation rates at 649 °C were improved marginally, but by a factor of no greater than 2; this finding is consistent with recent work on boron-doped IN718, which has demonstrated that the tortuosity of the fatigue crack path was greatly increased as the level of boron doping was increased from 12 ppm to 29 ppm; see Figure 4.36 [34]. The effect on fatigue initiation as manifested in the low-cycle-fatigue (LCF) life is very much more significant. Jain *et al.* [35] have doped Udimet 720Li in P/M form with varying concentrations of boron to 0.04 wt%, at a constant concentration of 0.025 wt%C; the effects of increasing the Zr concentration from 0.035 to 0.070 wt% and Hf from zero to 0.75 wt% were also studied. Testing was carried out at both 425 °C and 650 °C, at two grain sizes corresponding to ASTM 11 and ASTM 9, which were generated by sub- and super-solvus heat treatment, respectively. Un-doped Udimet 720Li in the cast-and-wrought condition was used as a control. The results are given in Figure 4.37 and Table 4.3. The results demonstrate that additions of the grain-boundary elements are beneficial to LCF life; improvements of at least 50% were noted for most conditions considered. Interestingly, the fatigue life for the fine-grained material (ASTM 11) was improved more substantially at 425 °C, whereas that for the coarse-grained material improved at 650 °C; no explanation of this finding was offered. In this study, no substantial

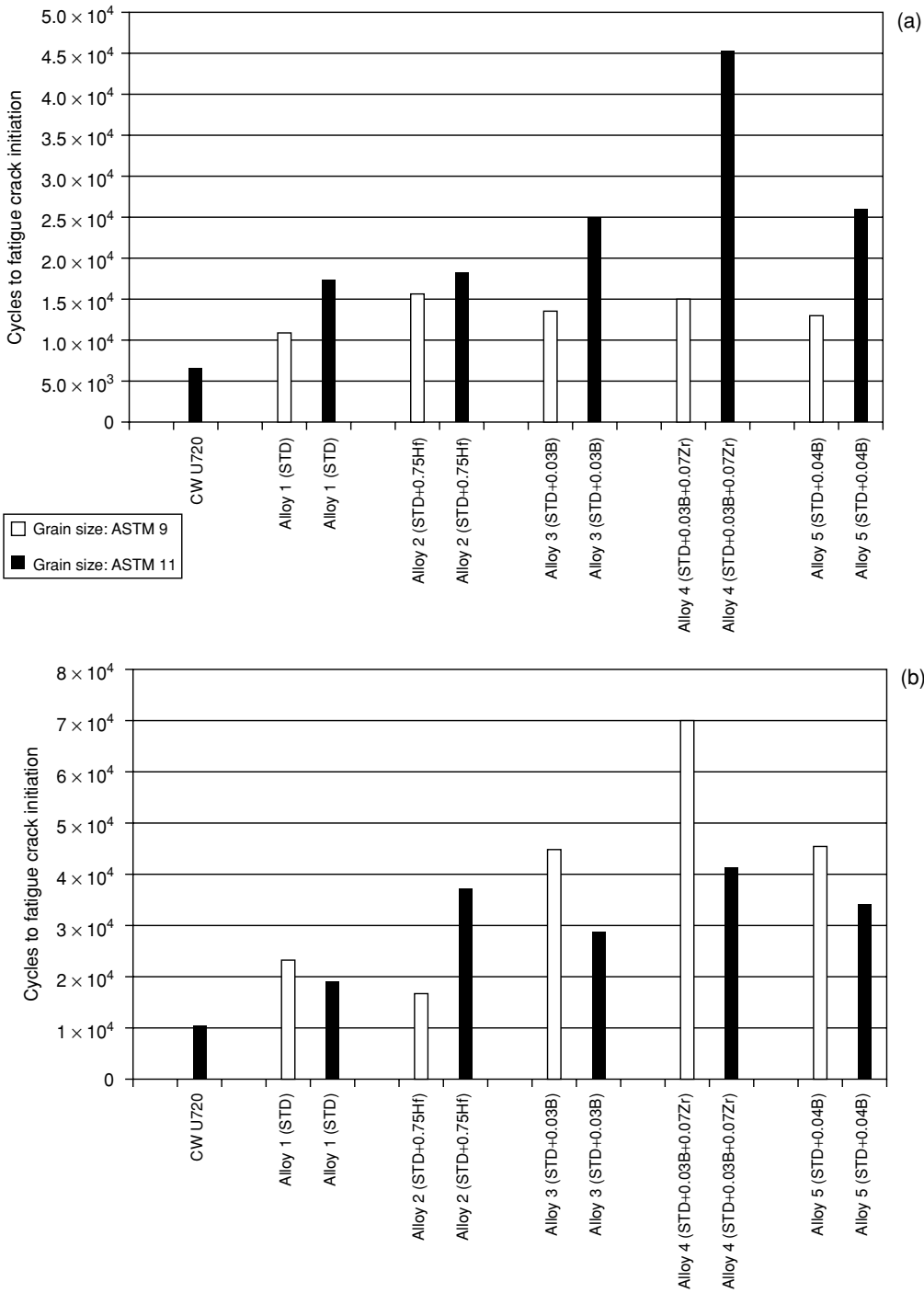


Fig. 4.37. Data showing the influence of small amounts of B, Zr and Hf on the LCF performance of the P/M U720Li turbine disc alloy [35]. LCF data at (a) 425 °C, $\Delta c_r = 0.9\%$, $R = 0.0$, air; (b) 650 °C, $\Delta c_r = 0.8\%$, $R = 0.0$, air. Note that CW U720 refers to cast-and-wrought material. For alloy compositions see Table 4.3.

improvements in tensile properties were caused by the additions of the grain-boundary strengtheners.

Blavette and co-workers have begun to unravel the scientific principles which govern the effects conferred by the grain-boundary elements [36]. Using a combination of field-ion microscopy and atom probe tomography, the characterisation of B, C and Zr grain-boundary chemistry in the P/M alloy N18 has been successfully attempted; see Figure 4.38. In this alloy, the bulk compositions of B, C and Zr are 0.083 at%, 0.075 at% and 0.018 at%, respectively, but substantial segregation of B and Mo was demonstrated at the γ/γ' grain boundaries, where the local concentrations were enhanced ten-fold compared to the bulk – equal to about 0.5 equivalent atomic monolayers. Incoherent γ/γ' boundaries were found to be similarly affected. Interestingly, no segregation of C was identified; it is probable that in this alloy it resides primarily in chromium-rich $M_{23}C_6$ carbides. No segregation of Zr was found. Similar results were found in earlier work on Astroloy [37].

Case study. The design of turbine disc alloys

Most of the alloy development work for turbine disc superalloys has been carried out by the engine manufacturers such as GE Aircraft Engines, Pratt & Whitney, SNECMA and Rolls-Royce. The number of alloys which have been designed is relatively small, and due to its proprietary nature it is unlikely that all details of the work have been revealed in the open literature; nevertheless, some information has emerged. When designing the powder disc superalloy N18 in the mid-1980s, SNECMA placed considerable emphasis [38] on promoting damage tolerance so that alloys with low-dwell fatigue crack propagation (FCP) rates were preferred. It was noted that for many of the P/M alloys developed to that point, such as Rene 95, IN100, MERL 76 and Astroloy, crack growth rates scaled with the yield stress (see Figure 4.39); N18 was designed to fall away from this correlation, an order of magnitude improvement in FCP rates being desired. For this alloy, an Ti/Al ratio of unity, trace levels of B, C and Zr at 0.15, 0.15 and 0.30 wt%, respectively and a relatively high concentration of Mo (6.5 wt%) were chosen [38]; it was claimed that Mo was more favourable than W because of its greater γ/γ' partitioning ratio and because it reduced notch sensitivity during creep at 650 °C. The lack of damage tolerance displayed by Rene 95 was acknowledged by GE in designing their Rene 88DT superalloy (note 'DT' refers to damage tolerant); significant improvements in these properties were made, with the ultimate tensile strength (UTS) of Rene 88DT being kept within 90% of that of Rene 95 [39]. Dwell fatigue crack propagation rates at 650 °C were improved by increasing the γ' fraction and size, increasing the Ti/Al ratio and Ta substitution for W; however a trade-off was identified which saw the high-volume-fraction γ' alloys exhibiting very good dwell FCP rates at 650 °C but poorer performance at 400 °C in the absence of a dwell period. Compared with Rene 95, the Rene 88DT composition was chosen to yield a smaller γ' fraction (40% rather than 50%), a higher Ti/Al ratio (1.8 rather than 0.7 on a wt% basis) and higher Cr and Co contents (16 wt% rather than 13 wt%, and 13 wt% rather than 8 wt%, respectively) – this last change being made despite substitutions of Co and Cr providing no effect on dwell-time fatigue crack growth rate. The significant effect of Ta in reducing

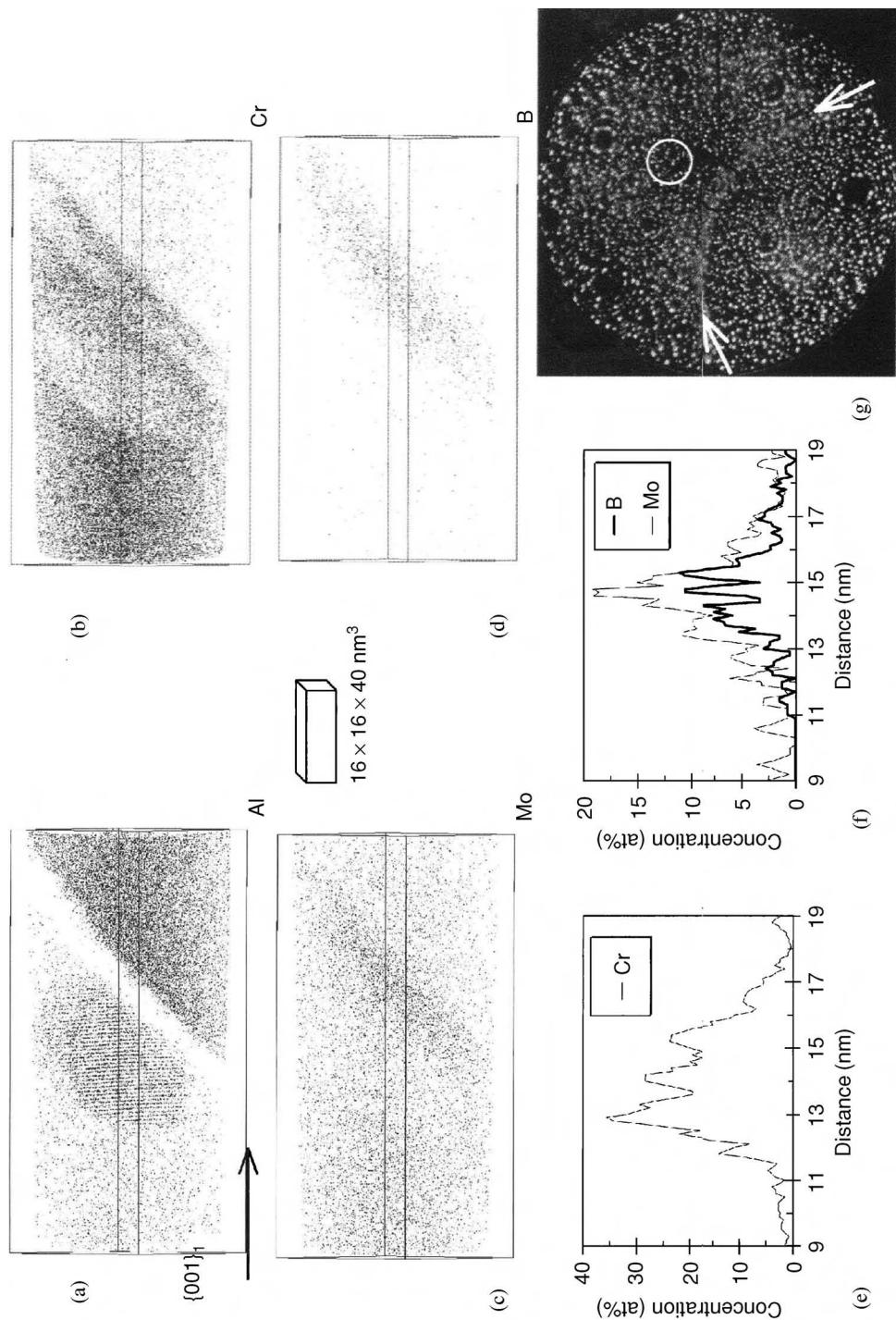


Fig. 4.38. Analysis using atom probe tomography of a γ/γ' grain boundary in N18 [36]: (a), (b), (c) and (d) correspond to maps of Al, Cr, Mo and B; (e) and (f) are concentration profiles normal to the boundary and (g) is a field-ion image (FIM) of the grain boundary.

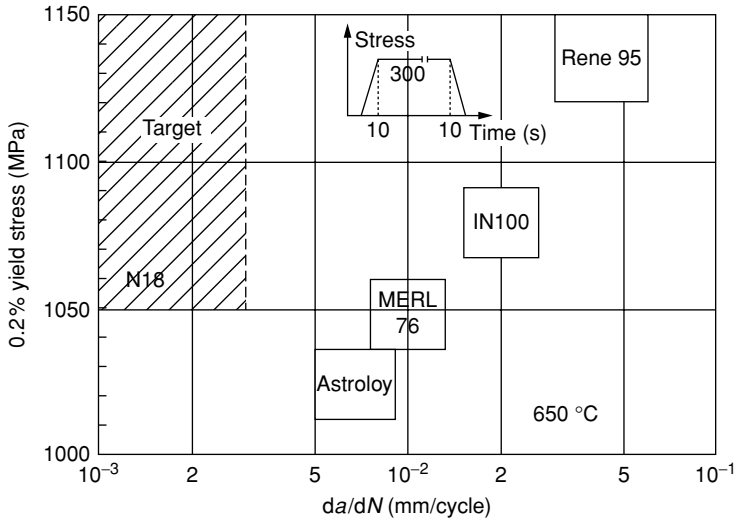


Fig. 4.39. Variation of the yield stress with dwell fatigue crack growth rate for a number of P/M superalloys [38].

dwell FCP rates has been noted in collaborative work by Pratt & Whitney, GE and NASA, which was aimed at developing an alloy for the High Speed Civil Transport (HSCT) aeroplane [40]; this gave rise to the alloy ME3, which has a significant Co concentration to lower the γ' solvus temperature so that the risk of thermally induced porosity (TIP) is minimised, with the heat treatment window being widened to enable super-solvus heat treatment and dual microstructure heat treatments; see Table 4.1. Alloy ME3 is similar to Rene 104, which will find use in the General Electric's GEnx engine for the Boeing 787 Dreamliner. An addition of Ta to Rolls-Royce's RR1000 powder disc superalloy was made for similar reasons as for ME3 [41], although the concentration was limited to 2.15 wt% since the FCP rates increased for higher concentrations of this element. Chromium was found to reduce the rate of FCP at elevated temperature, but its concentration was limited to 15 wt% because it promotes the precipitation of the σ phase, particularly above 700 °C. The importance of restricting the Cr concentration was learned by Special Metals Corporation during the Udimet 720 development programme [16] – the original compositional variant contained 18 wt%Cr, but in developing the second variant, Udimet 720Li, this was reduced to 16 wt%Cr, which has been shown to improve the susceptibility to σ formation [42]; moreover, the boron and carbon levels were reduced (Li refers to 'low interstitial') to about 0.025 wt% and 0.018 wt%, respectively, concentrations which appear to have been agreed upon by the superalloy community in at least an approximate sense; see Table 4.1. More recently, NASA has developed a hybrid disc alloy, LSHR, which combines the low solvus temperature of Rene 104 with the higher refractory content of Alloy 10, the latter being designed by Honeywell Engines and Systems to produce superior tensile and creep capability in small engines for auxiliary power unit (APU) applications; this was achieved by using a high refractory element content, most notably of tungsten.

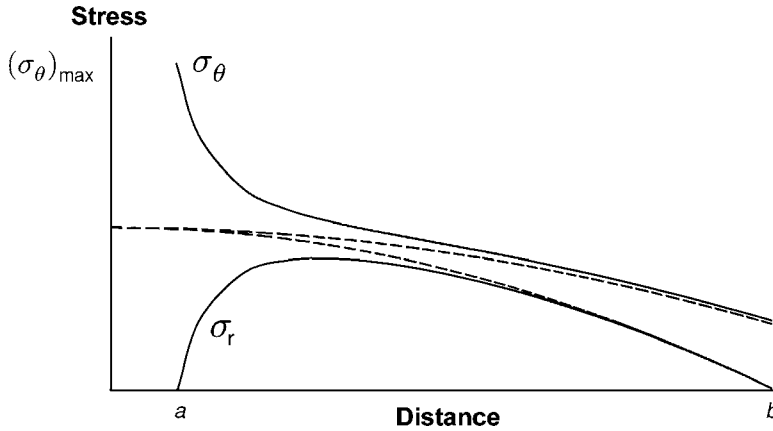


Fig. 4.40. Illustration of the stress field (solid lines) in a uniform disc of outer radius b , with a hole of radius a ; σ_θ and σ_r denote the hoop and radial stress, respectively. Also shown (dashed lines) is the behaviour of σ_θ and σ_r for the case when $a = 0$, i.e. when no hole exists.

4.3 Service life estimation for turbine disc applications

4.3.1 Stress analysis of a turbine disc of simplified geometry

A geometry of a typical turbine disc is complicated by the requirement for fixturing to accommodate the turbine blading, the drive arms necessary to transmit loads to the shaft, cover plates to manage the flow of cooling air and flanges with holes for the bolting together of neighbouring discs. Moreover, the shape of the disc is generally tapered to restrict the stress field at the bore to manageable levels. However, to gain a first appreciation of the very large stresses to which the material is subjected, one can consider a cylindrical component of outer diameter b containing a hole of inner radius a , rotating at an angular velocity ω . The density is ρ . Clearly the stress field will depend upon whether the component is thin (such that it approximates a disc) or thick (in which case it is a cylinder). Considering the situation for a disc under plane stress conditions – and given the simplifying assumptions – the radial and hoop components of the stress field, denoted σ_r and σ_θ , respectively, are given by [43,44]

$$\sigma_r = \frac{3 + \nu}{8} \rho \omega^2 \left(b^2 + a^2 - \frac{a^2 b^2}{r^2} - r^2 \right) \quad (4.2)$$

and

$$\sigma_\theta = \frac{3 + \nu}{8} \rho \omega^2 \left(b^2 + a^2 + \frac{a^2 b^2}{r^2} - \left[\frac{1 + 3\nu}{3 + \nu} \right] r^2 \right) \quad (4.3)$$

where the distance r is measured outwards from the axis of symmetry in a radial direction.

It is instructive to consider the stress field implied by these equations; see Figure 4.40. The hoop stress, σ_θ , is a maximum at the bore where $r = a$, where it takes a value of

$$(\sigma_\theta)_{\max} = \left[\frac{3 + \nu}{4} \right] \rho \omega^2 b^2 + \left[\frac{1 - \nu}{4} \right] \rho \omega^2 a^2 \quad (4.4)$$

Furthermore, σ_θ increases monotonically with decreasing r , due to the mass of material between $r = a$ and $r = b$ which requires support. The radial stress, σ_r , falls to zero at the free surfaces corresponding to the bore and rim, where $r = a$ and $r = b$, respectively, but rises to a maximum value between the two. That the hole provides a considerable stress concentration factor can be seen by setting $a = 0$ in Equations (4.2) and (4.3). Then, σ_θ and σ_r are equivalent at $r = 0$, when $\sigma_r = \sigma_\theta = (3 + \nu)\rho\omega^2 b^2/8$. A comparison with Equation (4.4) indicates that the stress concentration factor introduced by the hole is at least 2, and greater than this value if the second term in Equation (4.4) is significant.

Values for σ_r and σ_θ can be estimated. With $b = 33$ cm, $a = 7.5$ cm, $\omega = 10\,000$ rev/min, $\rho = 8900$ kg/m³ and $\nu = 0.3$, the hoop stress, σ_θ , has a maximum value of about 885 MPa at the bore where $r = a$. The stresses are very much greater for larger discs – Equation (4.4) indicates that for constant aspect ratio b/a , the bore stress scales as the square of the linear dimensions; consequently, larger engines must spin at a proportionately smaller rate. For example, if the linear dimensions are doubled at constant b/a , then ω must be halved for an equivalent stress field. These considerations confirm that materials with high-yield stresses are required for turbine-disc applications, and that the geometry needs to be chosen carefully such that the stress levels are restricted to tolerable levels. However, the level of conservatism in component design cannot be too great, otherwise excessive parasitic weight will be introduced.

4.3.2 *Methods for lifing a turbine disc*

The manufacture of a turbine disc calls for great care to be taken to avoid the introduction of defects. To see why this is the case, it is instructive to consider three common lifing methods.

A The life-to-first-crack approach

In the life-to-first-crack approach, which is used in the United Kingdom, current requirements stipulate that fracture critical components such as an HP turbine disc are treated according to a safe-life policy in which component ‘failure’ is defined as the occurrence of an engineering crack of surface length 0.75 mm [45,46]. This figure is below the length required for fast catastrophic fracture, and is chosen because it is about the limit of common non-destructive inspection (NDI) methods. Reliance is placed on the testing of real discs in spin rigs, under stress and temperature conditions comparable to those experienced in service. Each test can cost between \$250 000 and \$500 000; consequently, the number of tests is small – it is rare for this to exceed five.

For the purposes of declaring a safe service life, it is assumed that (i) the number of cycles to failure is distributed according to a log-normal density function, and (ii) to a 95% confidence level, not more than 1 in 750 discs are expected to contain a crack of surface length greater than 0.75 mm [47,48]; thus one is working at the 1/750 quantile (-3σ) such that the ratio of the fatigue lives at the $+3\sigma$ and -3σ points is 6; see Figure 4.41. Given this assumed scatter factor, the life corresponding to the lower 1/750 quartile is located at a factor of $\sqrt{6}$ below the geometric mean obtained from the spin testing. This lower

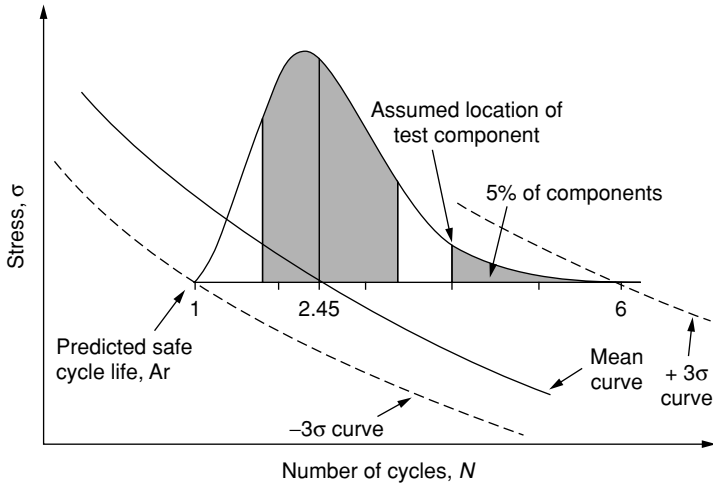


Fig. 4.41. Distribution of fatigue failure events superimposed on a $\sigma-N$ plot with the mean, $+3\sigma$ and -3σ values identified [47].

confidence bound translates to a factor of

$$6^{1.645/6\sqrt{n}} \quad (4.5)$$

where the 95% confidence limit corresponds to 1.645 standard deviations. This procedure allows a so-called 'safe life' to be calculated from the expression

$$\text{safe life} = \frac{(\prod_{i=1}^n N_i)^{1/n}}{2.449 \times 6^{(\frac{1.645}{6\sqrt{n}})}} \quad (4.6)$$

The denominator in the above equation, i.e. the factor by which the 'safe life' is lower than the geometric mean of the results from spin testing, is 4.00 when $n = 1$, falling to 3.25 when $n = 3$ and 2.90 when $n = 8$ – thus, a safety factor is present, the magnitude dependent upon the number of components tested.

Note that the use of the 'first crack' as the basis for the calculation will be conservative, since the length of a surface crack which will burst the disc is usually likely to be in excess of 0.75 mm. This provides a further built-in margin of safety.

B Damage-tolerant lifing

In practice, the 'extra' safety factor arising from the choice of a 0.75 mm long crack as constituting failure creates a dilemma. Precisely how large is it? If it is too large the procedure will be wildly conservative. It could, however, be very small or non-existent. Thus, in some ways the life-to-first-crack approach is unsatisfactory.

The 'damage-tolerant' approach to disc lifing (known alternatively as the 'retirement for cause' method) represents an alternative strategy [47,48]. One starts by estimating the length of crack required for 'burst', i.e. for fast catastrophic fracture, by using a fracture mechanics approach. Next, the number of cycles is estimated for the fatigue crack to grow

Table 4.4. Crack length data [48] relating to fast fracture and fatigue crack propagation in turbine disc alloys, with the stress range, $\Delta\sigma$, assumed equal to the proof strength

Stress range $\Delta\sigma$ /MPa (assumed = 0.2% proof strength)	a_{NDI} /mm	$a(3300)$ /mm	$a(6700)$ /mm	$a_{\text{crit}}(10\,000)$ /mm
800	0.30	0.50	1.20	11.1
850	0.23	0.40	0.97	9.8
950	0.13	0.23	0.58	7.9
1000	0.10	0.18	0.46	7.1
1200	0.04	0.08	0.20	4.9

Note: In the last column, the crack length, a_{crit} , at fast fracture is given, assuming $Y = 0.67$ and K_{Ic} equal to $100 \text{ MPa m}^{1/2}$ and a stress equal to the proof strength. In the other columns are given the crack lengths at initiation (which is equivalent to the required non-destructive inspection limit a_{NDI}), one-third and two-thirds dysfunction, assuming that failure occurs at 10 000 cycles.

from the limit of the non-destructive inspection techniques, a_{NDI} , to the size required for burst, a_{crit} . Fatigue-crack propagation data in the form of da/dN data are required for this purpose, and this usually conforms to the usual Paris law expression

$$\frac{da}{dN} = A \Delta K^m \quad (4.7)$$

where ΔK is the range of stress intensity experienced by the crack. The exponent m is usually between 3 and 4 for superalloys. Upon inserting the expression for $\Delta K = Y \Delta\sigma \sqrt{\pi a}$, and after performing the appropriate integration, one has

$$N_0 = \frac{1}{A(m/2 - 1)Y^m \Delta\sigma^m \pi^{m/2}} \left[\frac{1}{a_{\text{NDI}}^{m/2-1}} - \frac{1}{a_{\text{crit}}^{m/2-1}} \right] \quad (4.8)$$

where N_0 is the number of cycles for failure.

It is instructive to plug in some numbers, following Knott [48]. For many of the polycrystalline nickel-based superalloys used for turbine disc applications, the Paris relationship is of the form

$$\frac{da}{dN} = 4 \times 10^{-12} \Delta K^{3.3} \quad (4.9)$$

where ΔK is expressed in $\text{MPa m}^{1/2}$ and da/dN is the growth increment per cycle expressed in metres. Assuming a semicircular edge crack, $Y = 0.67$. For most nickel-based superalloys, the fracture toughness, K_{Ic} , is about $100 \text{ MPa m}^{1/2}$; if we assume $\Delta\sigma$ is equal to 800 MPa, with failure occurring at the top of the cycle, one obtains $a_{\text{crit}} = 11 \text{ mm}$. Next, one can determine an appropriate limit for a_{NDI} if one requires the turbine disc to last for at least 10 000 cycles – a figure of 0.30 mm is found. This is not very much smaller than the value assumed for the ‘first crack’ in the life-to-first-crack approach. The values of a at 3300 and 6700 cycles, corresponding to one-third and two-thirds dysfunction, respectively, are 0.50 mm and 1.20 mm; see Table 4.4 [48]. Note that the majority of the fatigue crack growth occurs in the last few thousand cycles. In practice, one should note that estimates of the fatigue life made in this way are *usually* conservative, since no account has been taken of